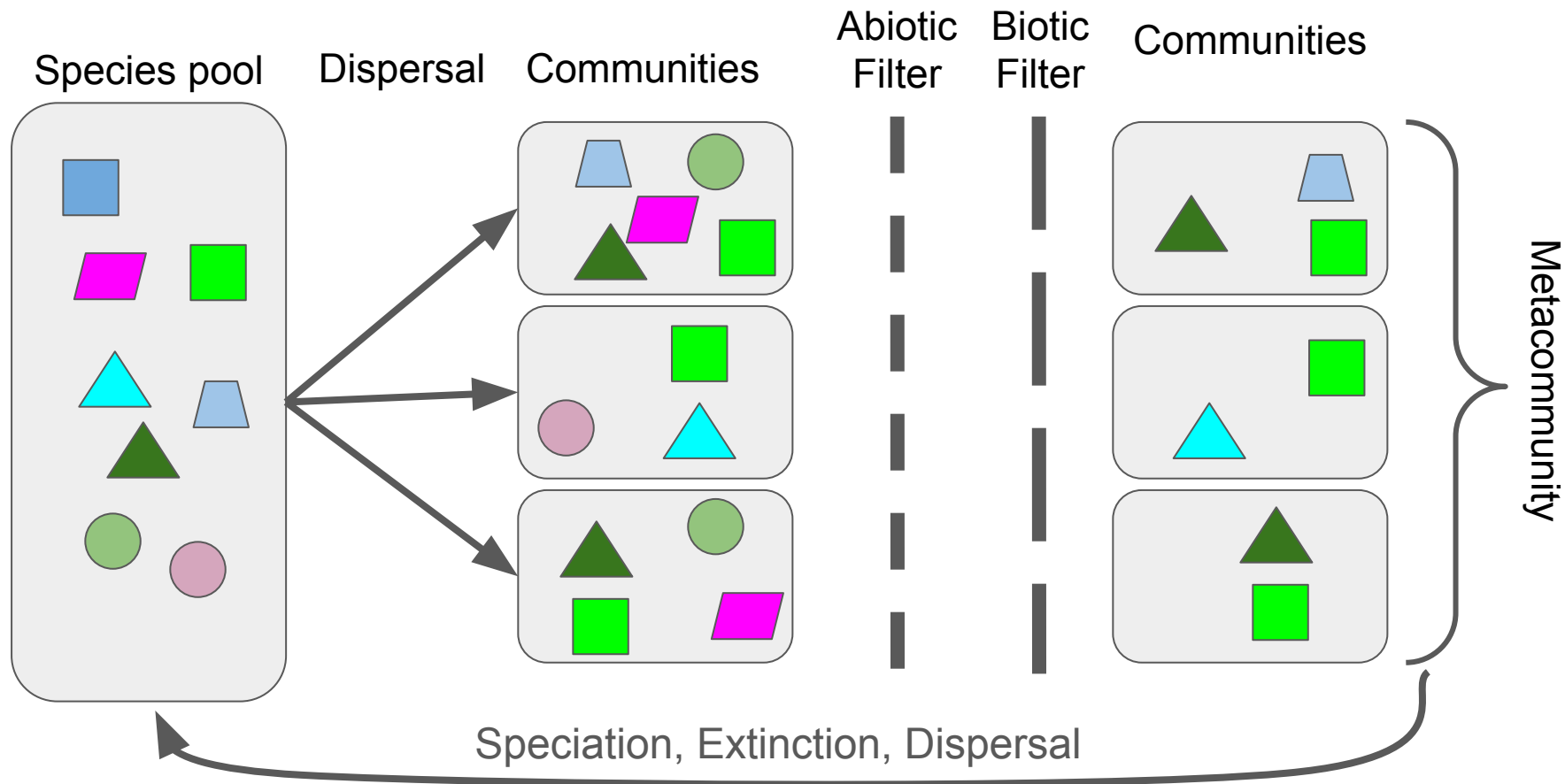


Community Assembly III

Today's Agenda:

- Quiz
- Community Assembly and Global Change



Community Assembly and Global Change

Focus on dispersal, biotic and abiotic filtering is well suited to global change:

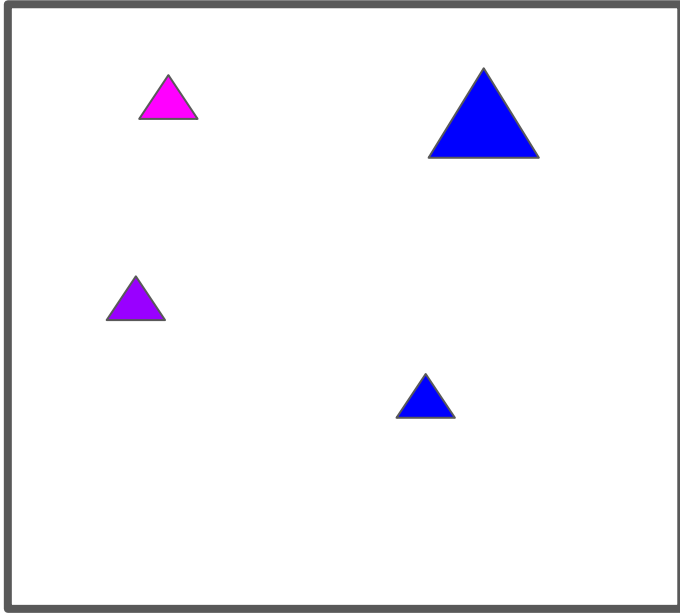
- How will climate or land use change impact communities? (abiotic)
- How will new species (rewilding/introduction) impact communities? (biotic)
- How will species respond to climate change? (dispersal, abiotic, biotic)

Examples for today

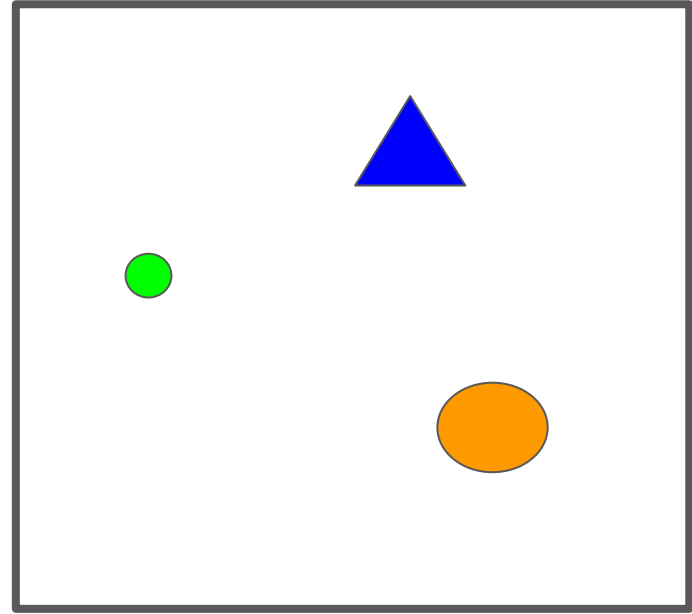
- 1) Changing bird communities due to introductions
- 2) Changing communities due to climate change

Quantifying Diversity

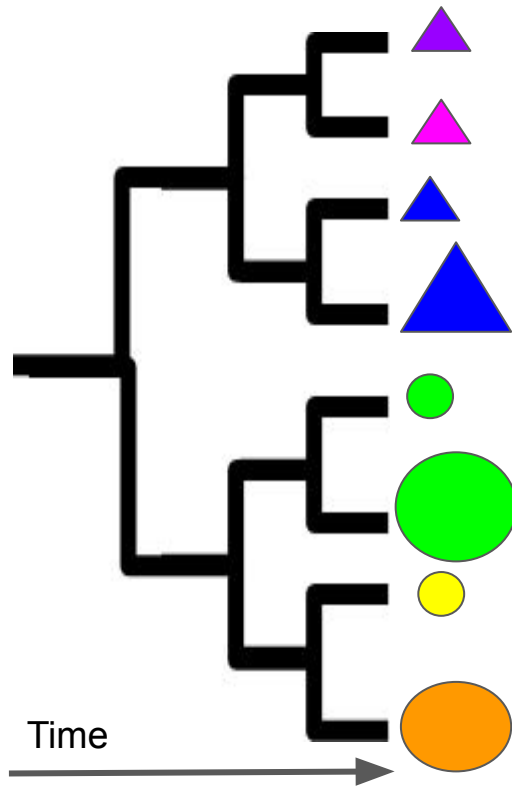
Community A



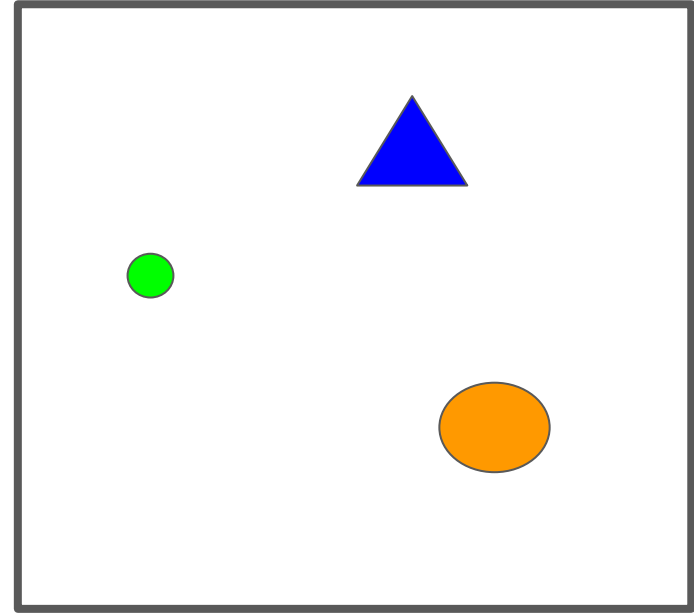
Community B



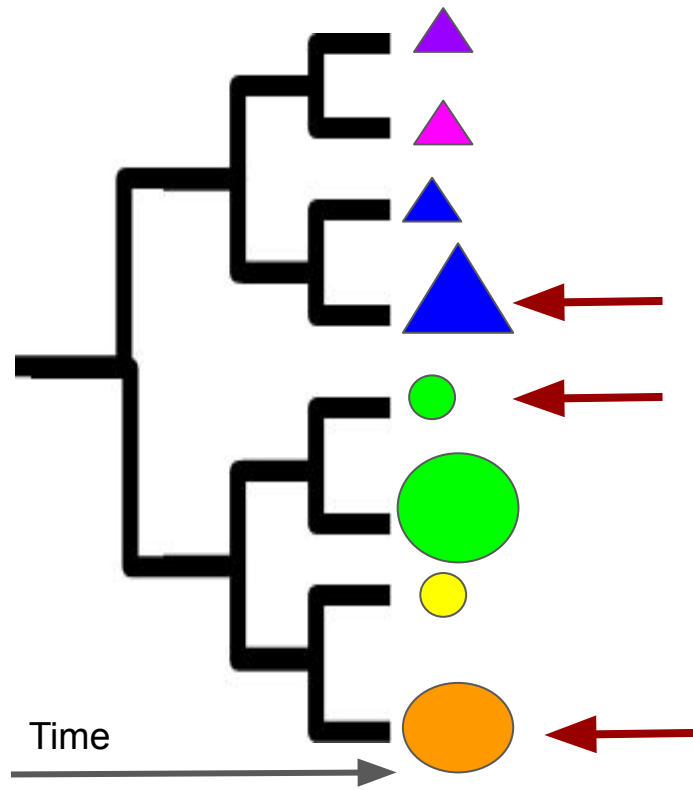
Quantifying Diversity: Community Phylogenetics



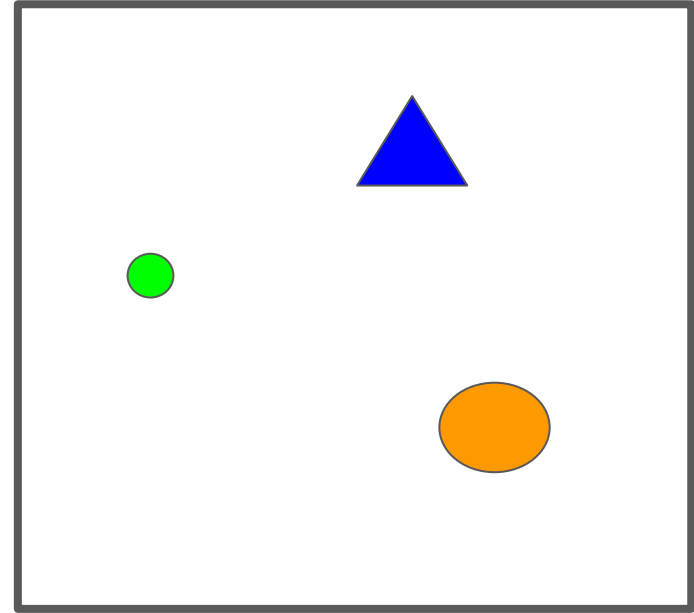
Community B



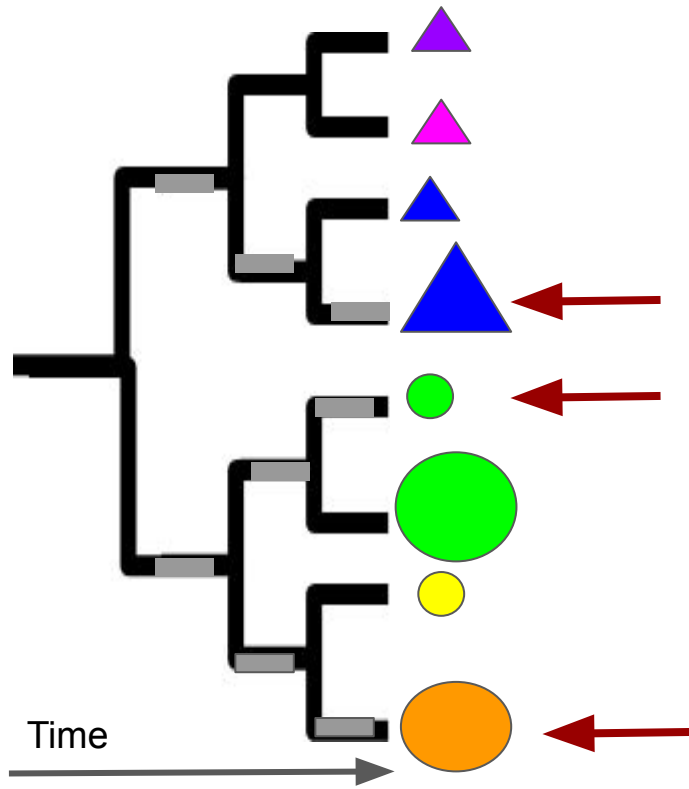
Quantifying Diversity: Community Phylogenetics



Community B

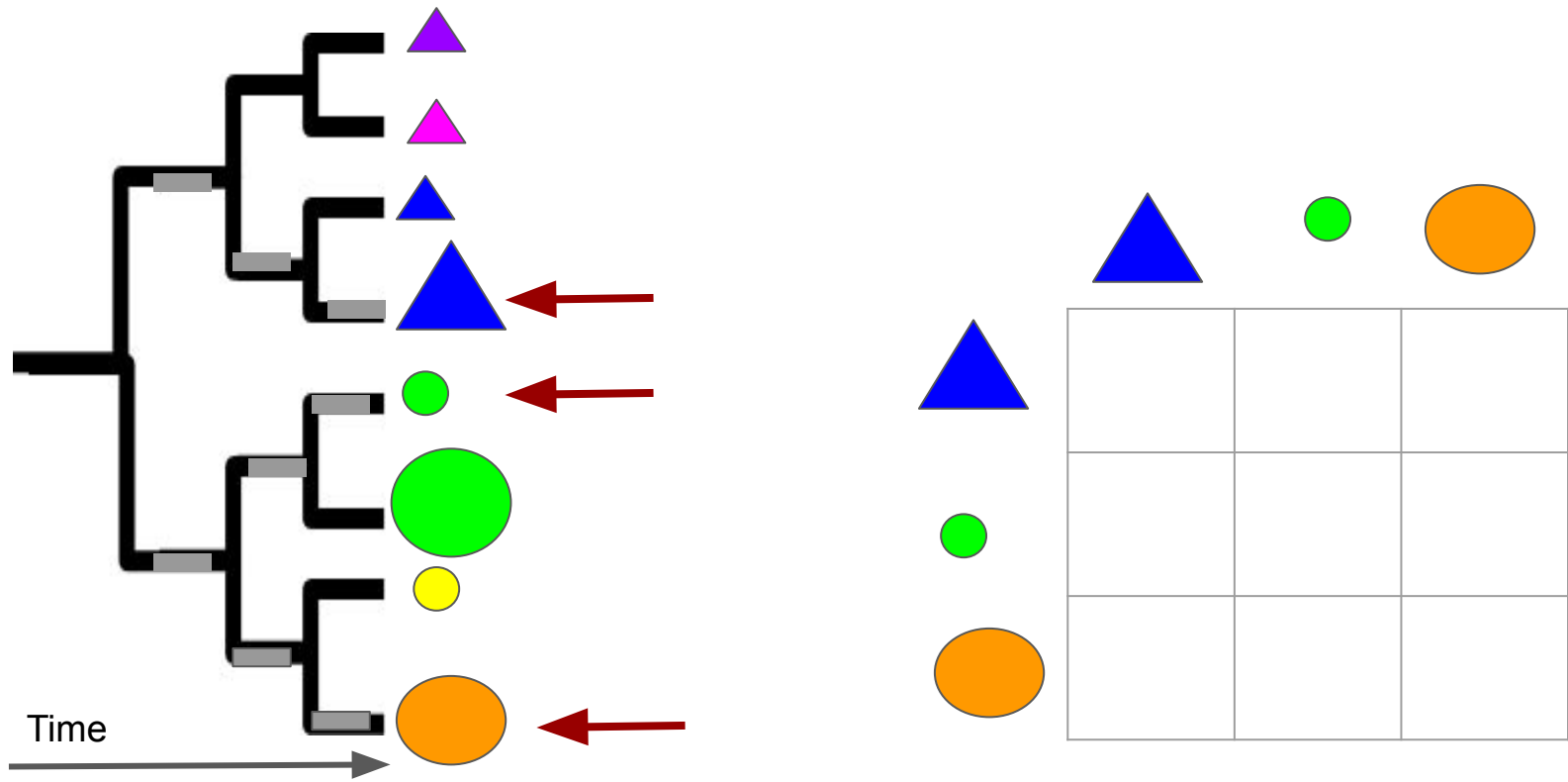


Quantifying Diversity: Community Phylogenetics

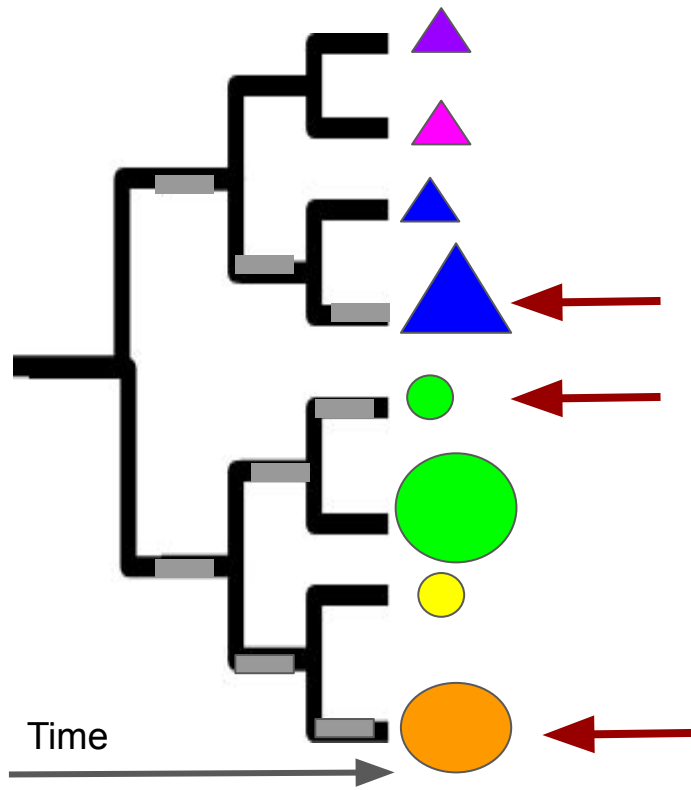


Phylogenetic Diversity = 8
Total branch length connecting species

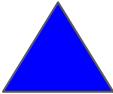
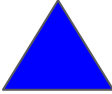
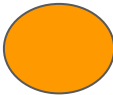
Quantifying Diversity: Community Phylogenetics



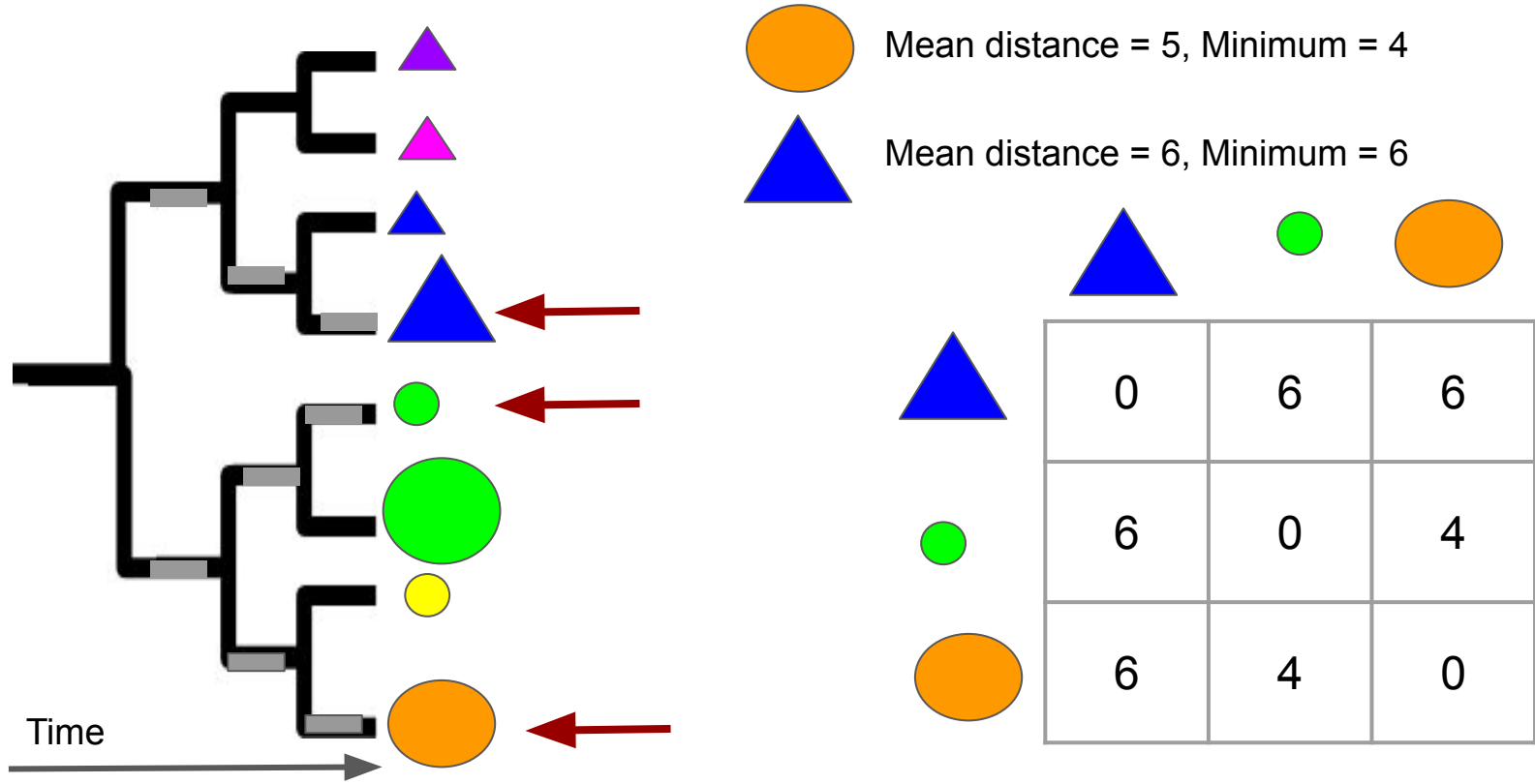
Quantifying Diversity: Community Phylogenetics



A distance matrix table showing the pairwise distances between three taxa: a blue triangle, a green circle, and an orange circle. The taxa are listed as headers for both rows and columns. The diagonal elements are 0, representing the distance from a taxon to itself. The off-diagonal elements represent the pairwise distances between the taxa.

			
	0	6	6
	6	0	4
	6	4	0

Quantifying Diversity: Community Phylogenetics



Community Phylogenetics

RESEARCH ARTICLE

Phylogenetic structure of soil bacterial communities predicts ecosystem functioning

Eduardo Pérez-Valera, Marta Goberna and Miguel Verdú*

Ketola et al. *BMC Ecol* (2017) 17:15
DOI 10.1186/s12898-017-0126-z

BMC Ecology

RESEARCH ARTICLE

Open Access



Propagule pressure increase and phylogenetic diversity decrease community's susceptibility to invasion

T. Ketola^a, K. Saarinen and L. Lindström

esa

ECOSPHERE

Jointly modeling niche width and phylogenetic distance to explain species co-occurrence

TAMMY L. ELLIOTT^{1,†} AND T. JONATHAN DAVIES^{2,3}

LETTER

doi:10.1038/nature14372

Phylogenetic structure and host abundance drive disease pressure in communities

Ingrid M. Parker^{1,2*}, Megan Saunders³, Megan Bontrager¹, Andrew P. Weitz¹, Rebecca Hendricks³, Roger Magarey⁴, Karl Suiter⁴ & Gregory S. Gilbert^{2,3*}

ECOLOGY LETTERS

Ecology Letters, (2011) 14: 782–787

doi: 10.1111/j.1461-0248.2011.01644.x

LETTER

Phylogenetic limiting similarity and competitive exclusion

Abstract

Cyrille Violle,^{1,2,3*} Diana R. Nemergut,^{4,5} Zhichao Pu¹ and Lin Jiang¹

One of the oldest ecological hypotheses, proposed by Darwin, suggests that the struggle for existence is stronger between more closely related species. Despite its long history, the validity of this phylogenetic limiting similarity hypothesis has rarely been examined. Here we provided a formal experimental test of the hypothesis using pairs of bacterivorous protist species in a multigenerational experiment. Consistent with the hypothesis

Community Phylogenetics and Introductions

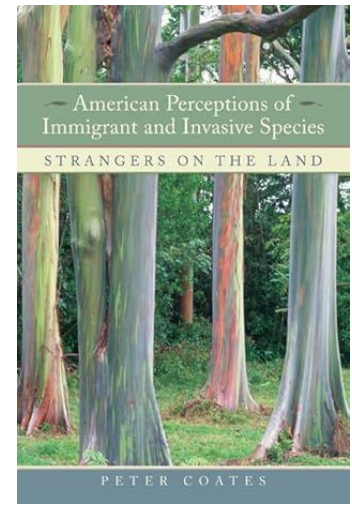
Biological introductions serve as “natural” experiments in community assembly

-Also of major economic and ecological interest



Quick aside: “Invasions” terminology

- The term invasions is often applied to biological introductions
- Confusing term
- Sometimes a legal term for unwanted introductions
- Sometimes used to describe rapidly spreading introductions
- Sometimes just means introduced by people
- Perceptions tied to other cultural or moral issues



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Comment | Published: 08 June 2011

Don't judge species on their origins

[Mark A. Davis](#) , [Matthew K. Chew](#) , [Richard J. Hobbs](#) , [Ariel E. Lugo](#) , [John J. Ewel](#) , [Geerat J. Vermeij](#) , [James H. Brown](#) , [Michael L. Rosenzweig](#) , [Mark R. Gardener](#) , [Scott P. Carroll](#) , [Ken Thompson](#) , [Steward T. A. Pickett](#) , [Juliet C. Stromberg](#) , [Peter Del Tredici](#) , [Katharine N. Suding](#) , [Joan G. Ehrenfeld](#) , [J. Philip Grime](#) , [Joseph Mascaro](#)  & [John C. Briggs](#) 

[Nature](#) **474**, 153–154 (2011) | [Cite this article](#)

50k Accesses | **735** Citations | **659** Altmetric | [Metrics](#)

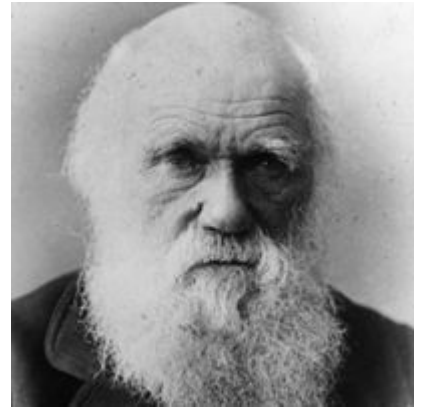
Conservationists should assess organisms on environmental impact rather than on whether they are natives, argue Mark Davis and 18 other ecologists.

Community Phylogenetics and Introductions

In “The Origin of Species”, Darwin proposed a hypothesis:

Successful introductions should be close relatives of natives

- close relatives share adaptations to the new environment



Community Phylogenetics and Introductions

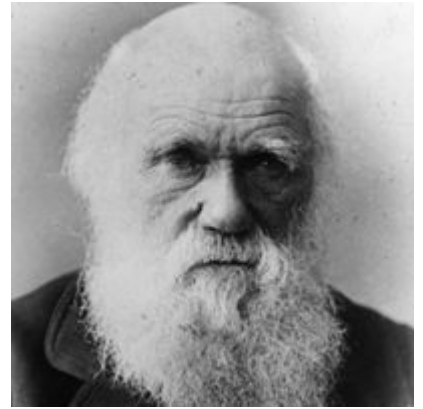
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Tested, and rejected, this hypothesis based on the work of DeCandolle

- Regions gain “far more in new genera than in new species”



Community Phylogenetics and Introductions

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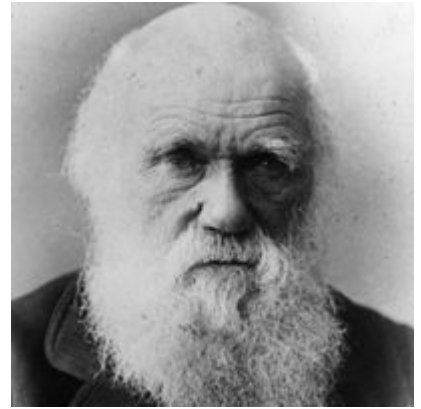
- close relatives share adaptations to the new environment

Tested, and rejected, this hypothesis based on the work of DeCandolle

- Regions gain “far more in new genera than in new species”

Darwin proposed a second hypothesis:

- competition between closely-related species limits success



Tests of Darwin's naturalization conundrum

PNAS

A test of Darwin's naturalization hypothesis in the thistle tribe shows that close relatives make bad neighbors

Daniel S. Park¹ and Daniel Potter

Department of Plant Sciences, University of California, Davis, CA 95616

Edited by Peter H. Raven, Missouri Botanical Garden, St. Louis, Missouri, and approved September 18, 2013 (received for review May 25, 2013)

Darwin's naturalization conundrum can be explained by spatial scale

Daniel S. Park^{a,b,1}, Xiao Feng^{c,d}, Brian S. Maitner^b, Kacey C. Ernst^e, and Brian J. Enquist^{b,f}



Ecography 35: 614–623, 2012

doi: 10.1111/j.1600-0587.2011.07176.x

© 2011 The Authors. Ecography © 2011 Nordic Society Oikos

Subject Editor: Catherine Graham. Accepted 29 July 2011

Patterns of bird invasion are consistent with environmental filtering

Brian S. Maitner, Jennifer A. Rudgers, Amy E. Dunham and Kenneth D. Whitney

B. S. Maitner (bsm2@rice.edu), J. A. Rudgers, A. E. Dunham and K. D. Whitney, Rice Univ., Dept of Ecology and Evolutionary Biology, Houston, TX 77005, USA.

PROCEEDINGS B

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Research

Cite this article: Ma C, Li SP, Pu Z, Tan J, Liu M, Zhou J, Li H, Jiang L. 2016 Different effects of invader – native phylogenetic relatedness on invasion success and impact: a meta-analysis of Darwin's naturalization hypothesis. *Proc. R. Soc. B* **283**: 20160663. <http://dx.doi.org/10.1098/rspb.2016.0663>

Different effects of invader – native phylogenetic relatedness on invasion success and impact: a meta-analysis of Darwin's naturalization hypothesis

Chao Ma^{1,2,3}, Shao-peng Li¹, Zhichao Pu¹, Jiaqi Tan¹, Manqiang Liu³, Jing Zhou⁴, Huixin Li³ and Lin Jiang¹

¹School of Resources and Environment, Anhui Agricultural University, Hefei 230036, People's Republic of China

²School of Biology, Georgia Institute of Technology, Atlanta, GA 30332, USA

³College of Resources and Environmental Sciences, Nanjing Agricultural University, Nanjing, Jiangsu 210095, People's Republic of China

⁴Institute of Soil Science, Chinese Academy of Sciences, Nanjing, Jiangsu 210008, People's Republic of China

ML, 0000-0001-6654-7795; LJ, 0000-0002-7114-0794

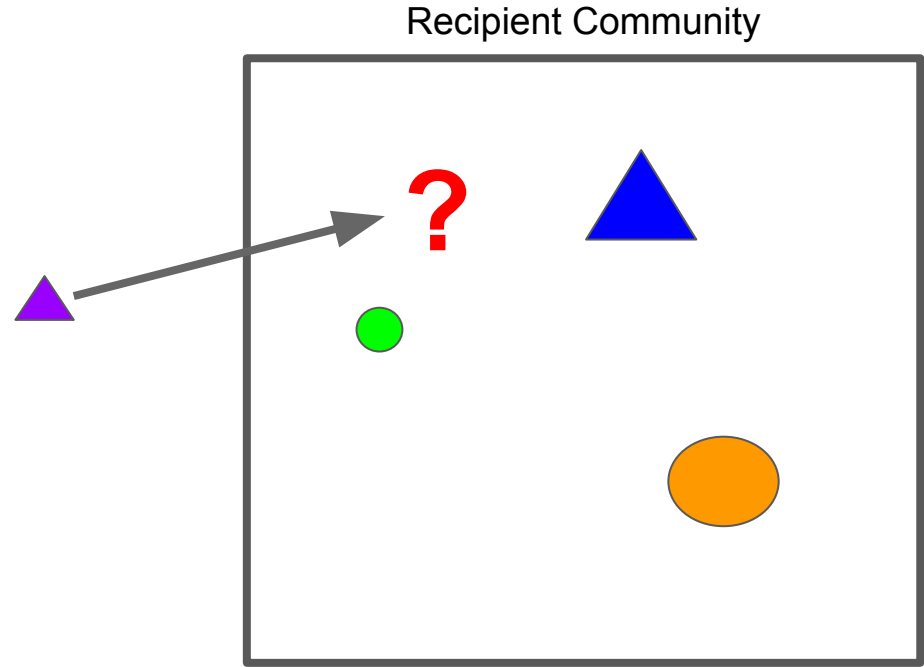
Diversity and Distributions, (*Diversity Distrib.*) (2016) **22**, 318–331



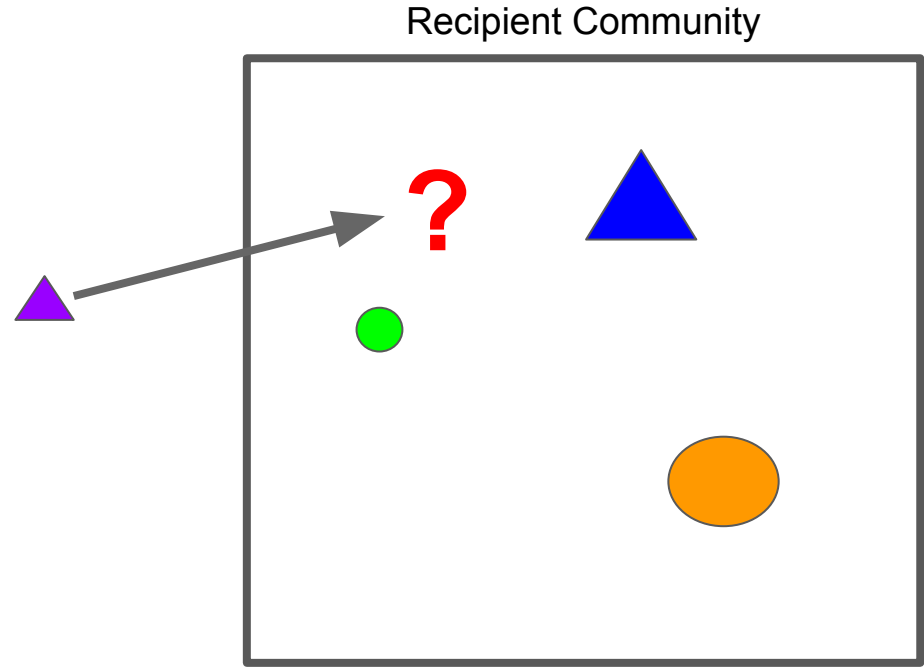
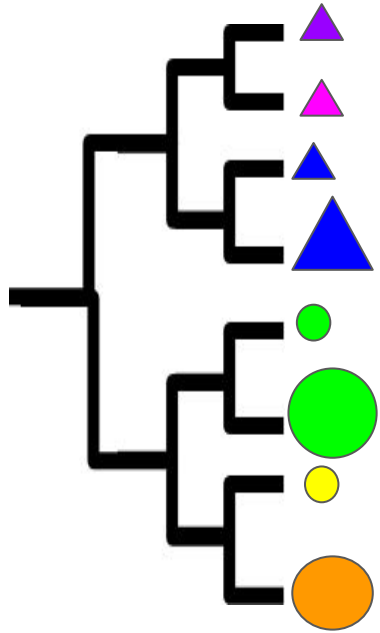
Deconstructing Darwin's Naturalization Conundrum in the San Juan Islands using community phylogenetics and functional traits

Hannah E. Marx^{1,*}, David E. Giblin², Peter W. Dunwiddie² and David C. Tank¹

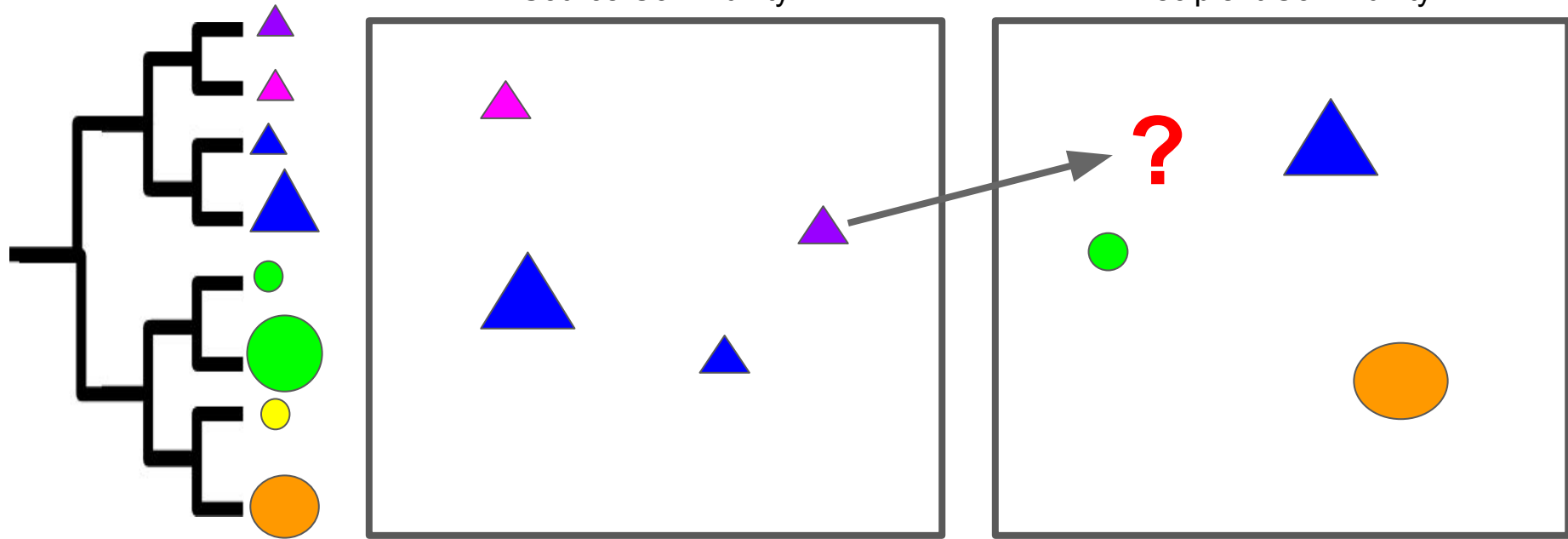
Tests of Darwin's naturalization conundrum



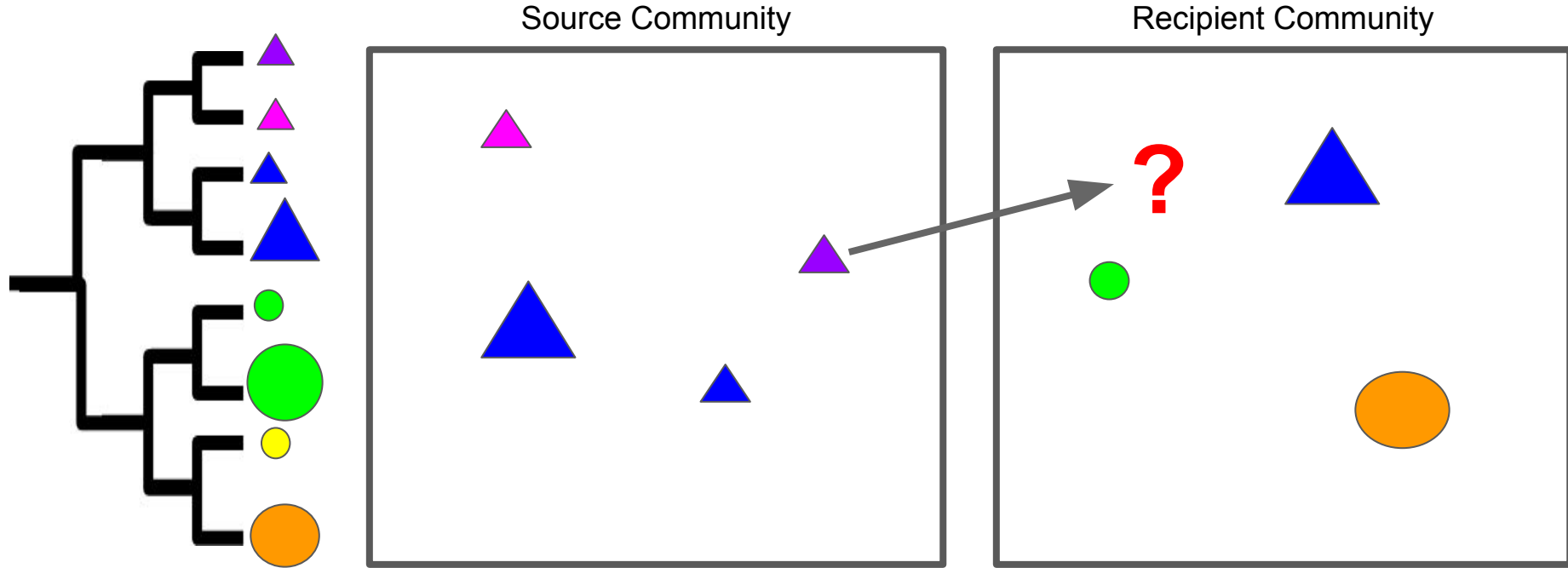
Tests of Darwin's naturalization conundrum



Tests of Darwin's naturalization conundrum

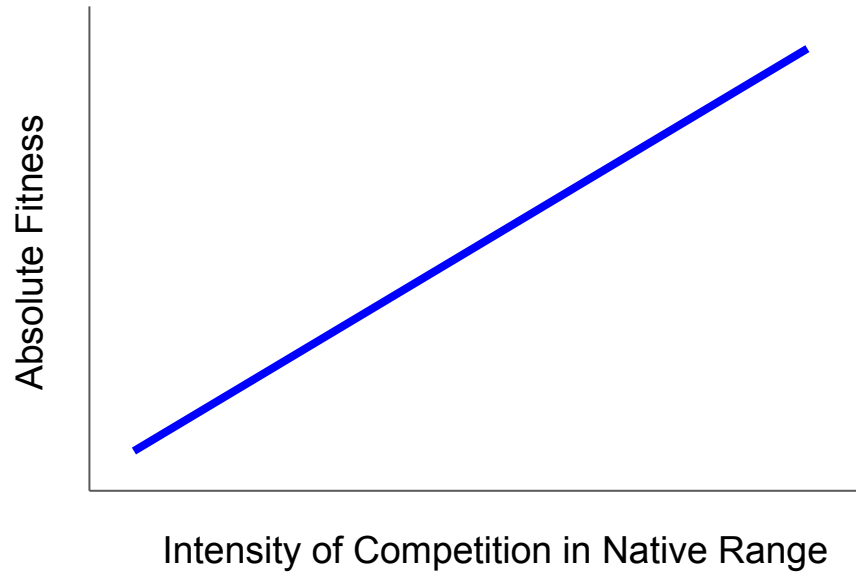
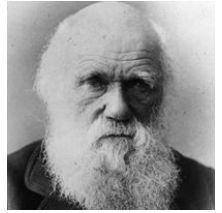


Tests of Darwin's naturalization conundrum



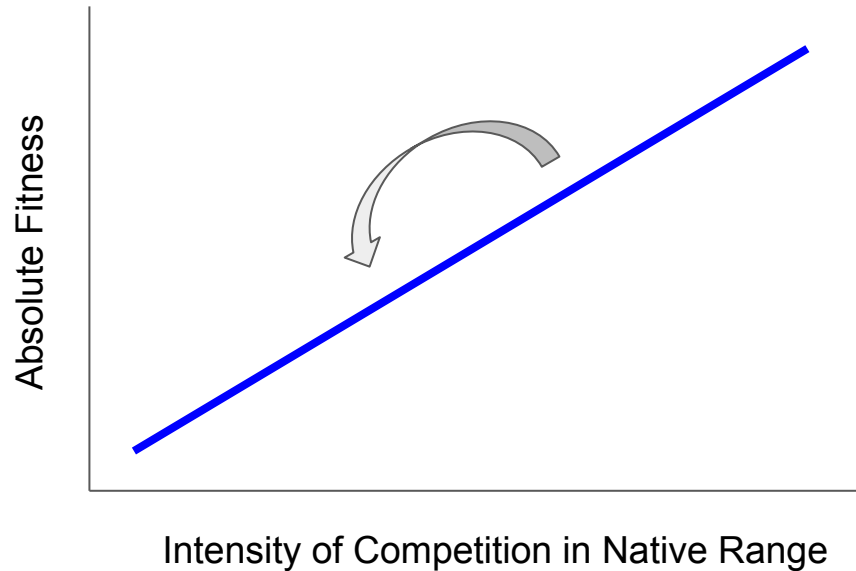
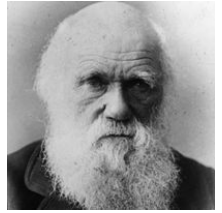
There are hypotheses as to why the SOURCE community might matter!

Evolutionary Imbalance Hypothesis



Higher competition selects for optimized resource use.

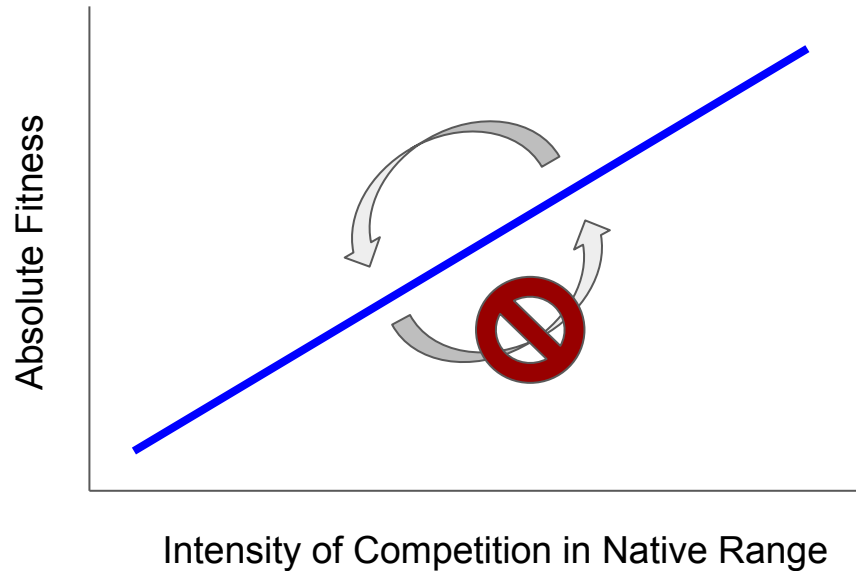
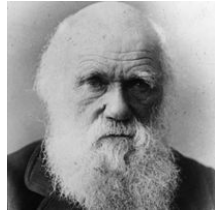
Evolutionary Imbalance Hypothesis



Higher competition selects for optimized resource use.

Species should be able to invade communities with less intense competition.

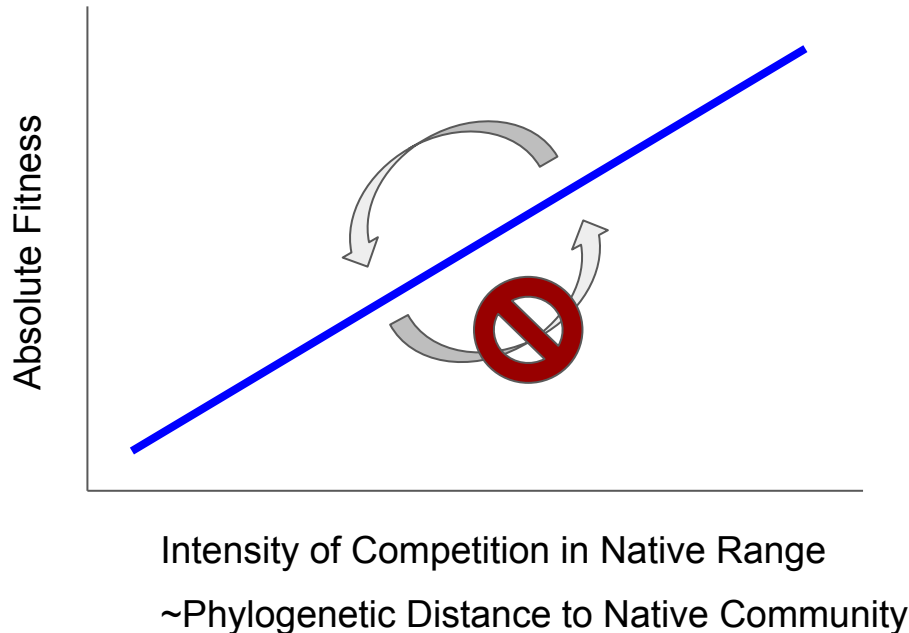
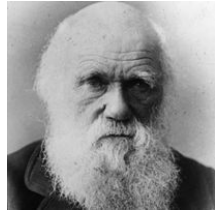
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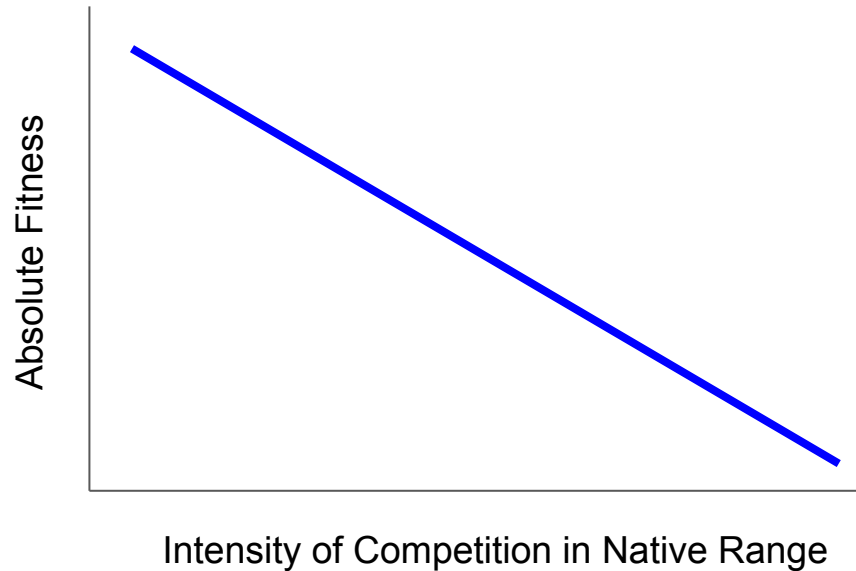


Higher competition selects for optimized resource use.

Species should be able to invade communities with less intense competition.

Species coming from regions with many, close relatives should be more successful

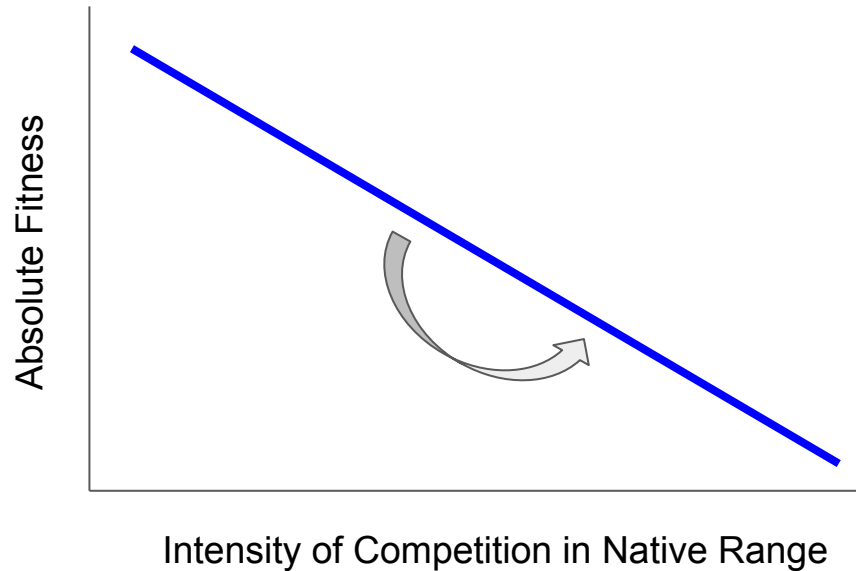
Competitive Constraint Hypothesis



Higher competition limits populations sizes
Fewer beneficial mutations
Reduced selection efficiency

Reduced ability to explore phenotype space

Competitive Constraint Hypothesis

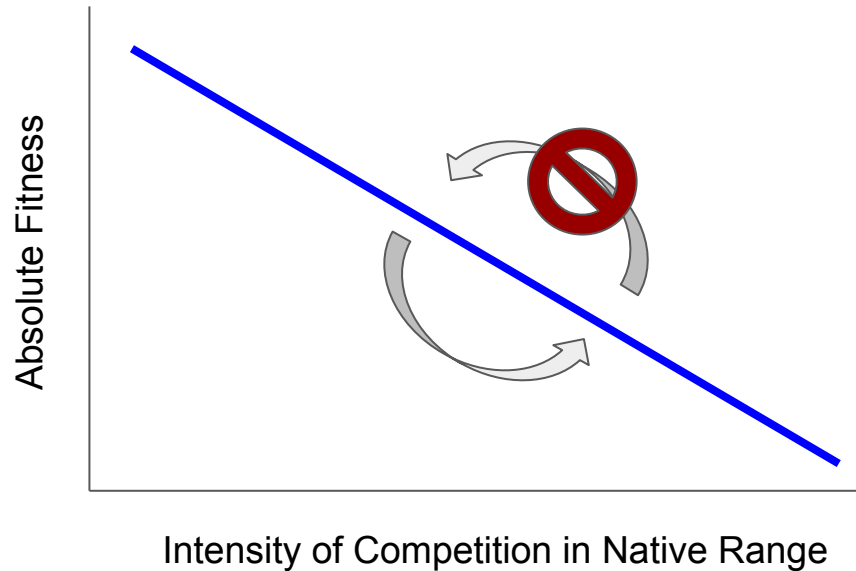


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Competitive Constraint Hypothesis

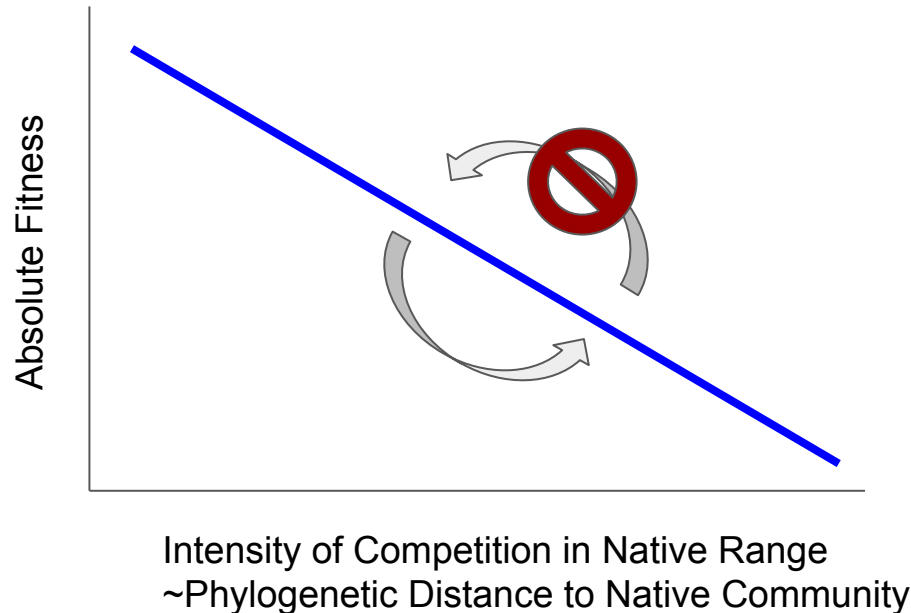


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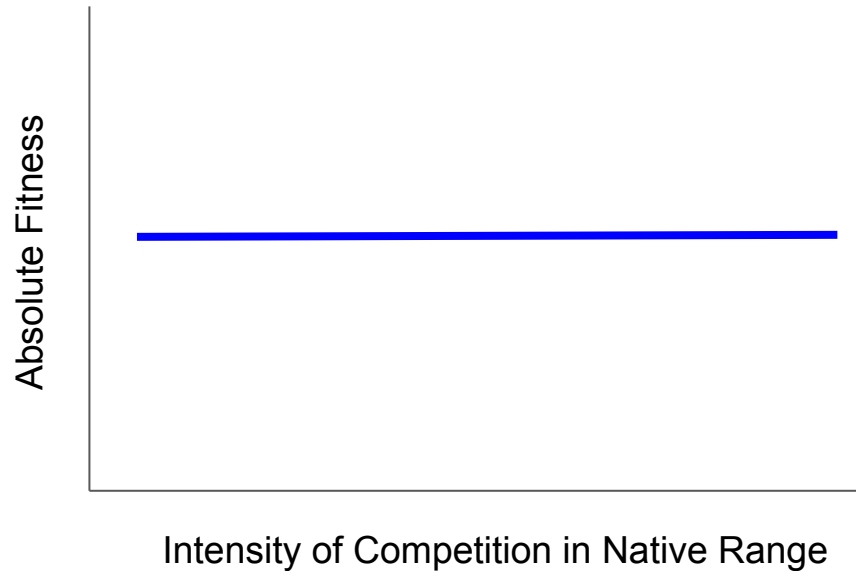
Higher competition limits populations sizes
Fewer beneficial mutations
Reduced selection efficiency

Reduced ability to explore phenotype space

Species should be able to invade communities with more intense competition.

Species coming from regions with few, distant relatives should be more successful

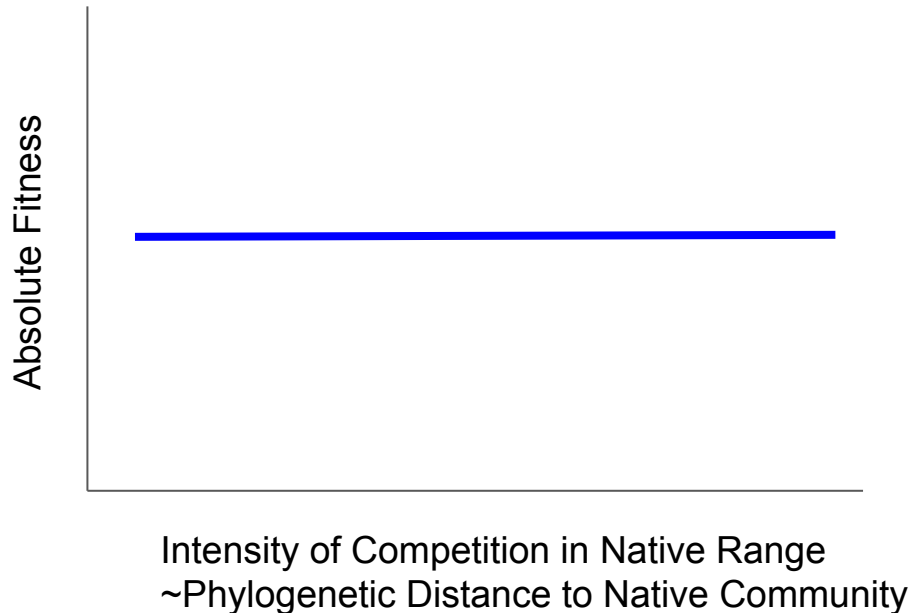
Universal Trade-Off Hypothesis



All species are constrained to similar trade-offs

Species from different regions won't differ systematically in competitive ability

Universal Trade-Off Hypothesis

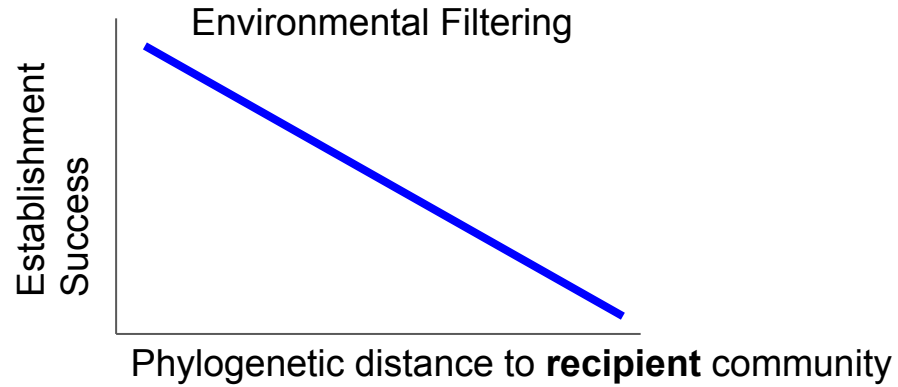


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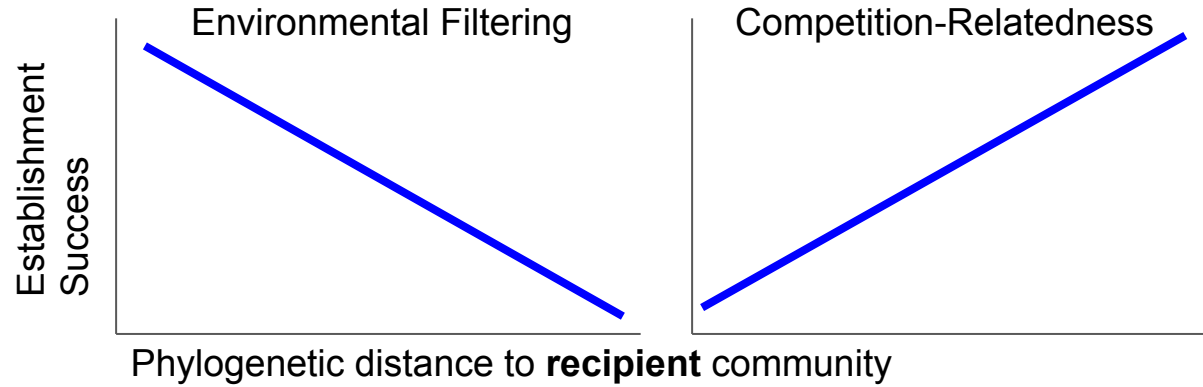
Species from different regions won't differ systematically in competitive ability

No relationship between source region phylogenetic structure and invasion success

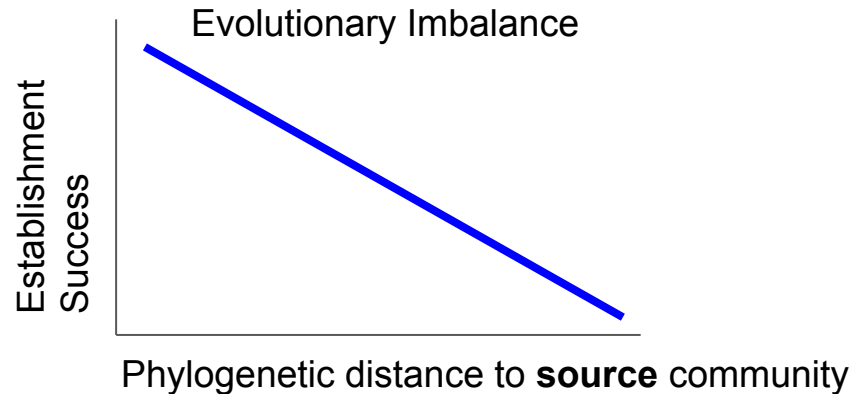
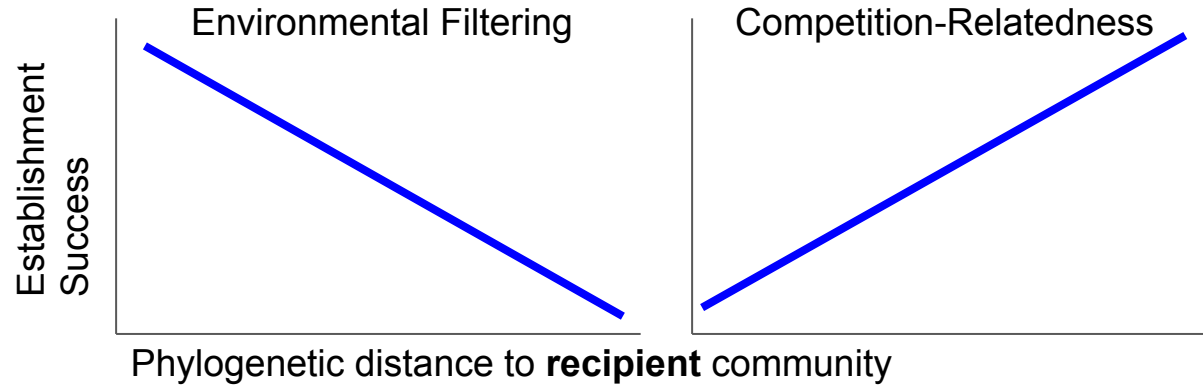
Establishment Hypotheses



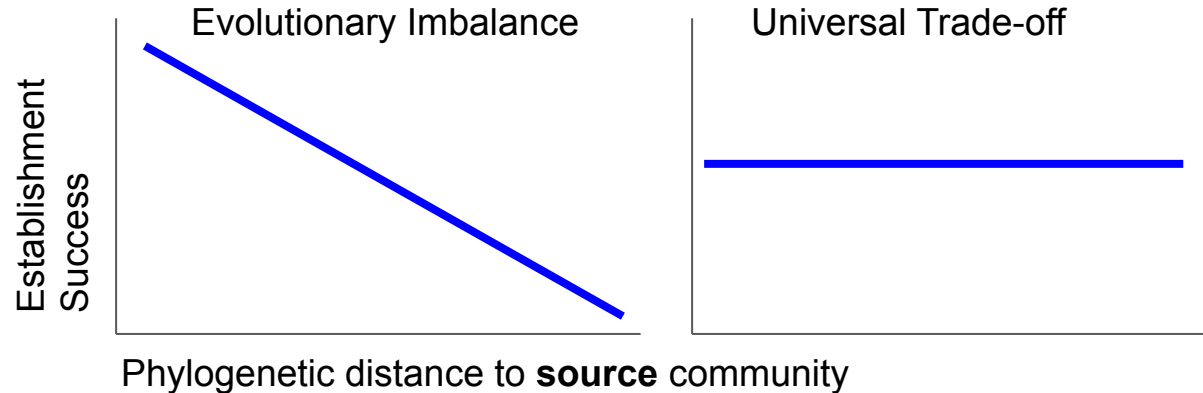
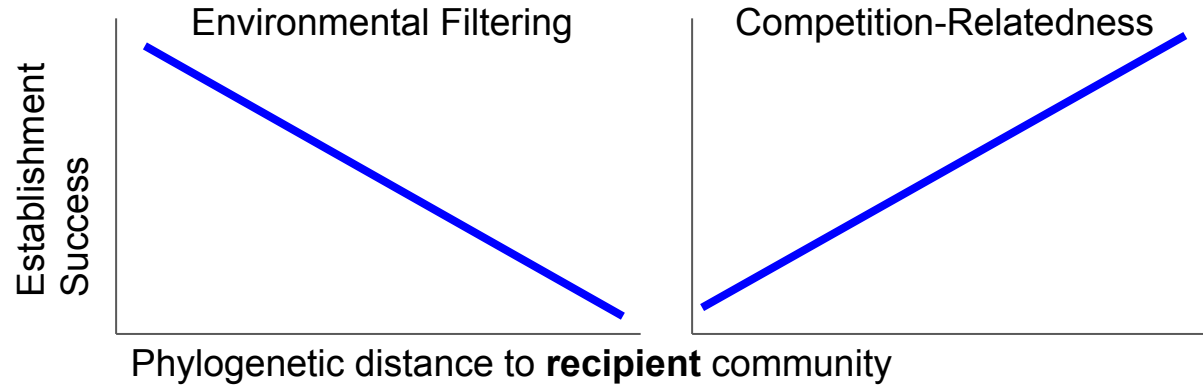
Establishment Hypotheses



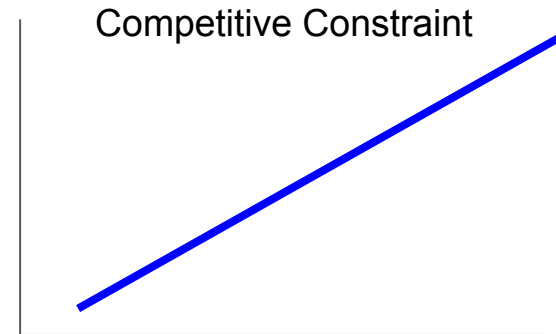
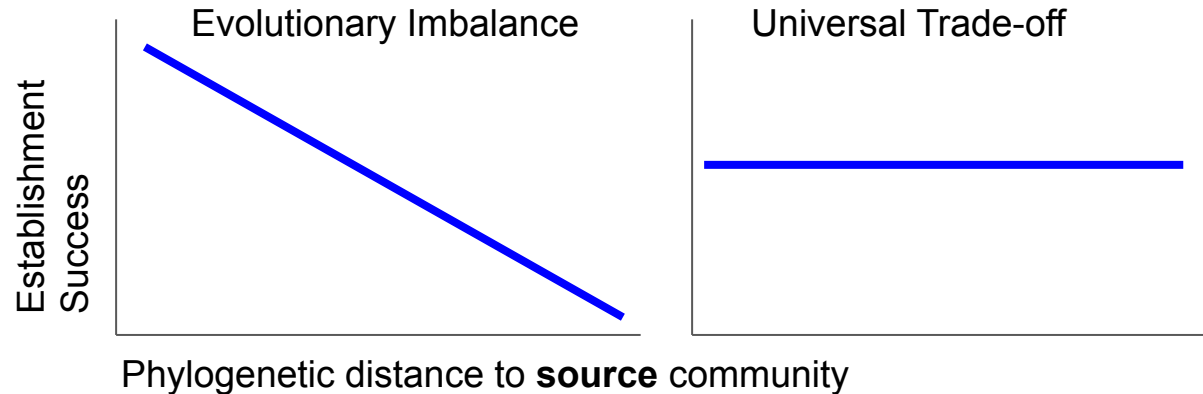
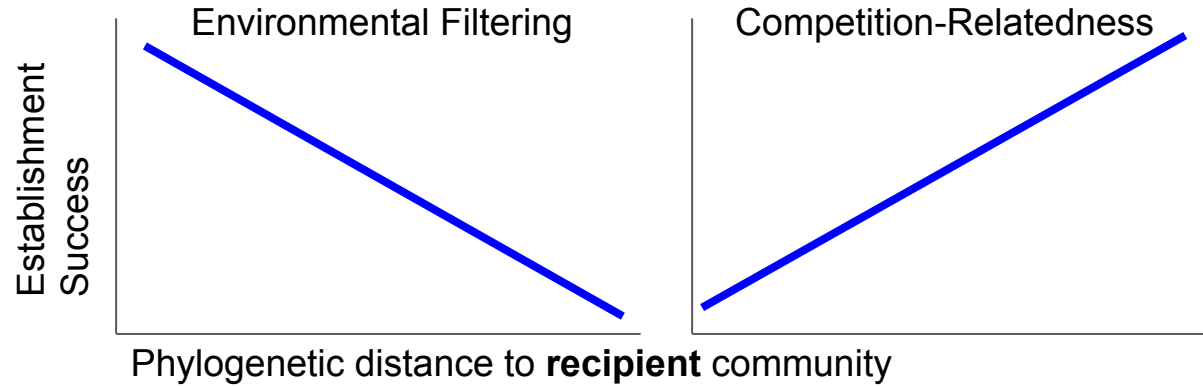
Establishment Hypotheses



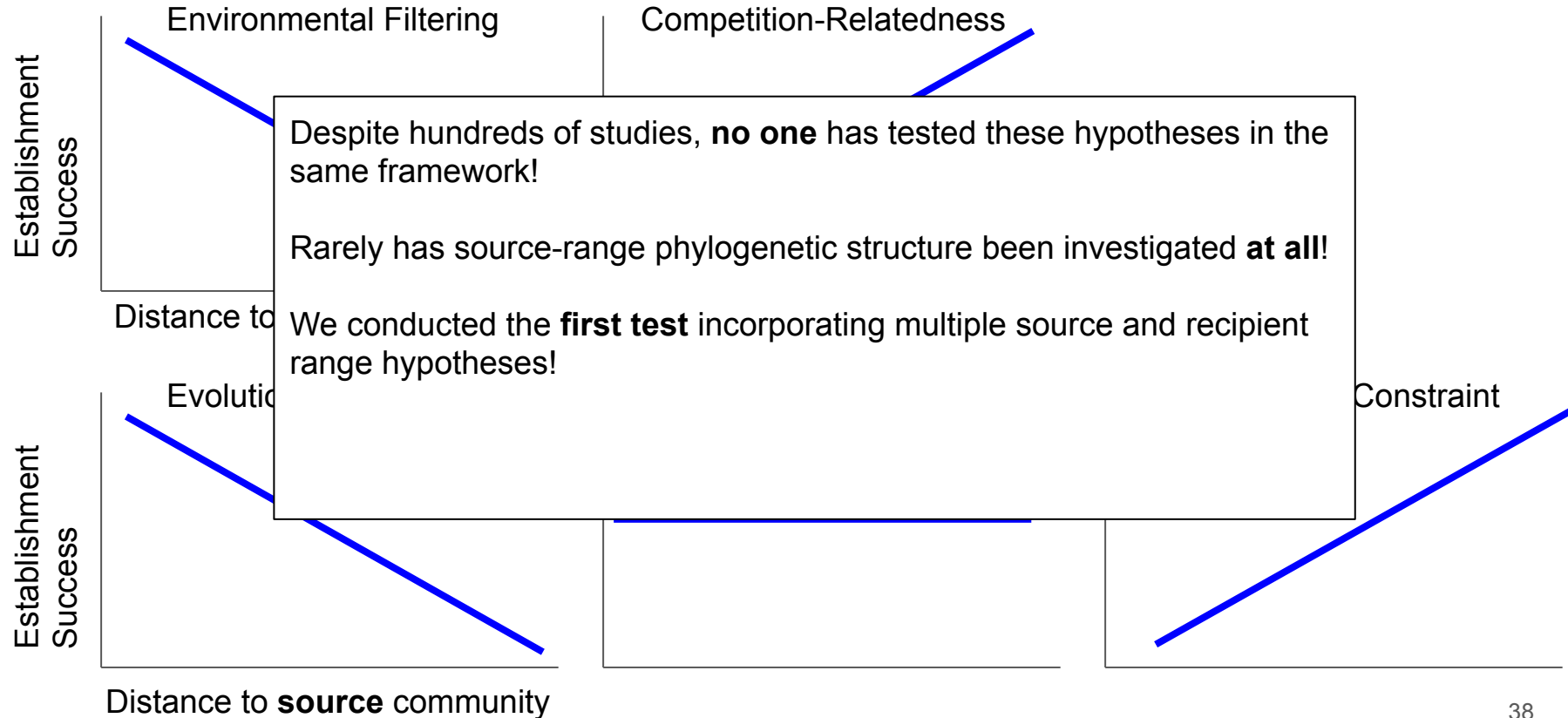
Establishment Hypotheses



Establishment Hypotheses



Establishment Hypotheses



Testing Establishment Hypotheses

In order to test these hypotheses we needed:

- Successful vs. failed introduction data

- Range maps for source and recipient regions

- A complete phylogeny

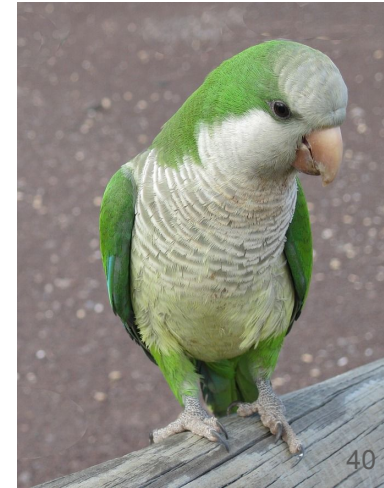
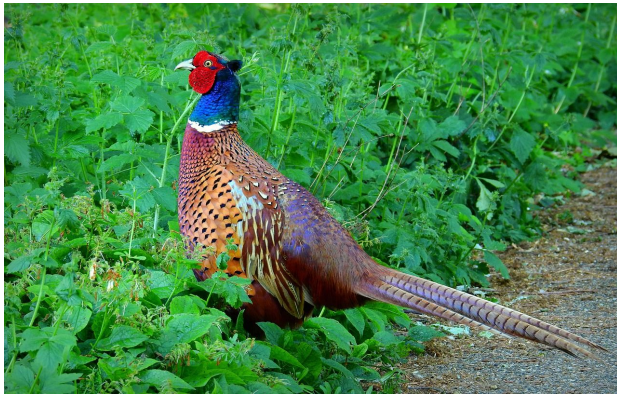
Testing Establishment Hypotheses

In order to test these hypotheses we needed:

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Phylogenies and establishment: Questions

Can source-range phylogenetic structure predict establishment success?

Does incorporating source-range phylogenetic structure improve models?

Which hypotheses are supported ?

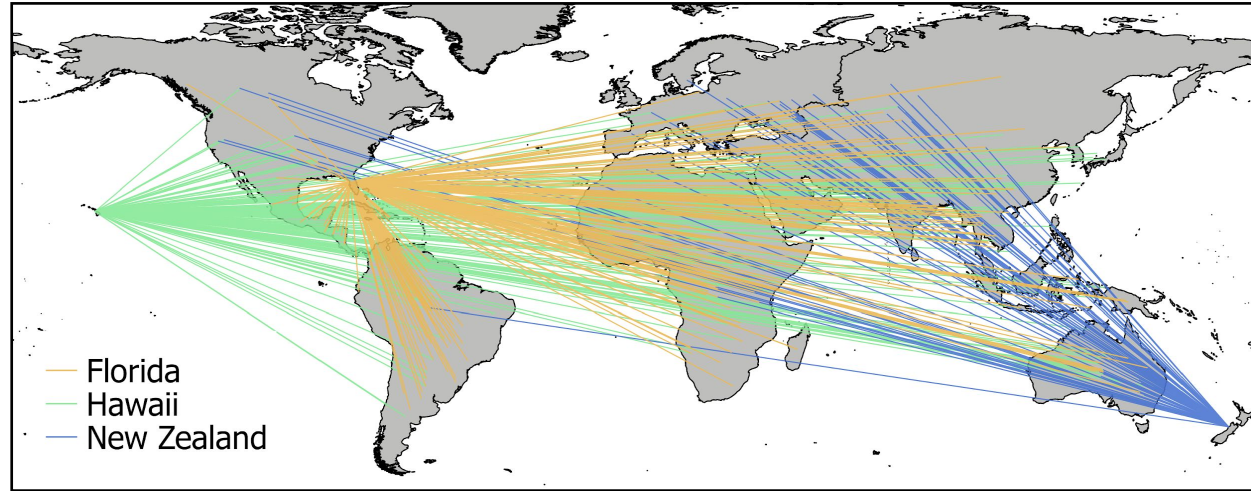
Phylogenies and establishment: Data

Community data

- Birdlife International

Introduction data:

- Moulton et al. 2001
- Ornithological Society of New Zealand 2003
- Pranty 2004



Phylogenies:

- Jetz et al. (2012)

Phylogenies and establishment: Methods

For each introduction, we calculated:

Source range:

Range size

Species richness

Phylogenetic distances between focal species, other species

Summarized as minimum, mean, variance, skewness, kurtosis

Total phylogenetic diversity within source range

Recipient range:

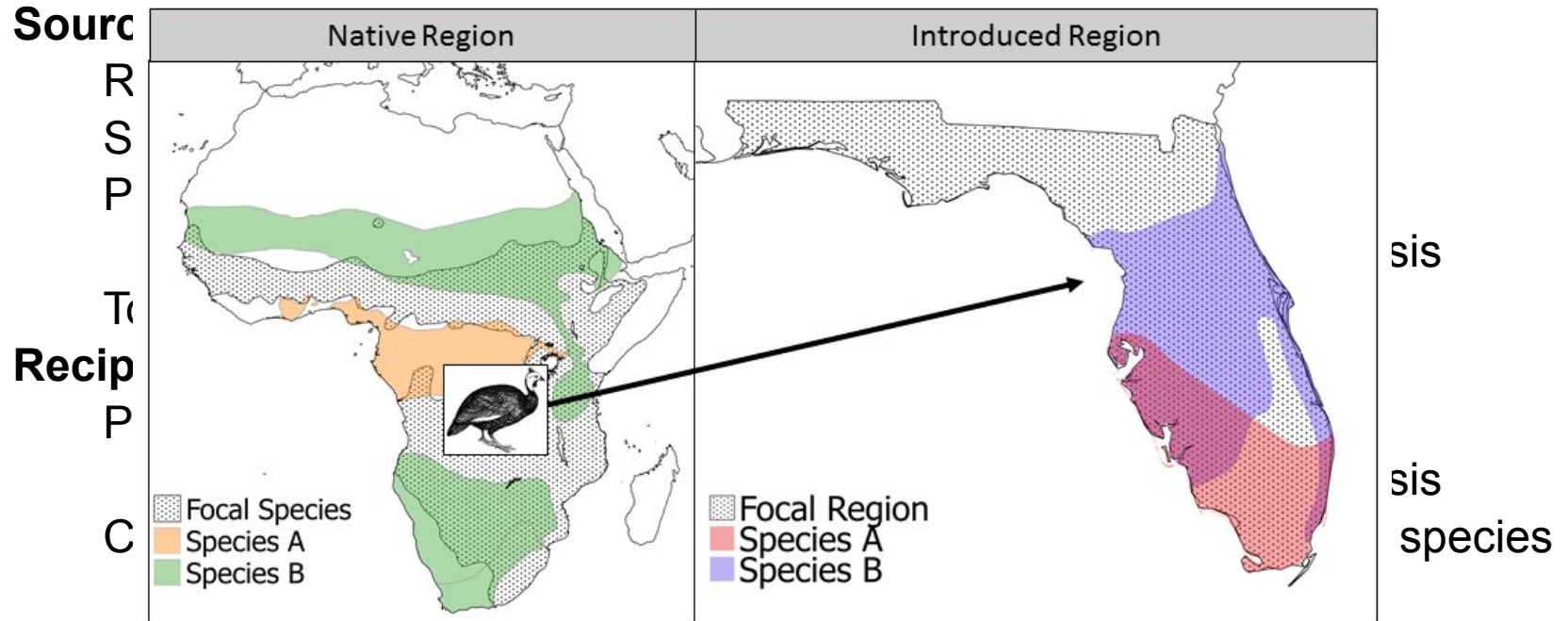
Phylogenetic distances between focal species, other species

Summarized as minimum, mean, variance, skewness, kurtosis

Calculated relative to native, established, native and established species

Phylogenies and establishment: Methods

For each introduction, we calculated:



Analyses: Does Source Region Matter?

**Establishment success ~ source region phylogenetic structure +
source region species richness +
source region area + (1|recipient region)**

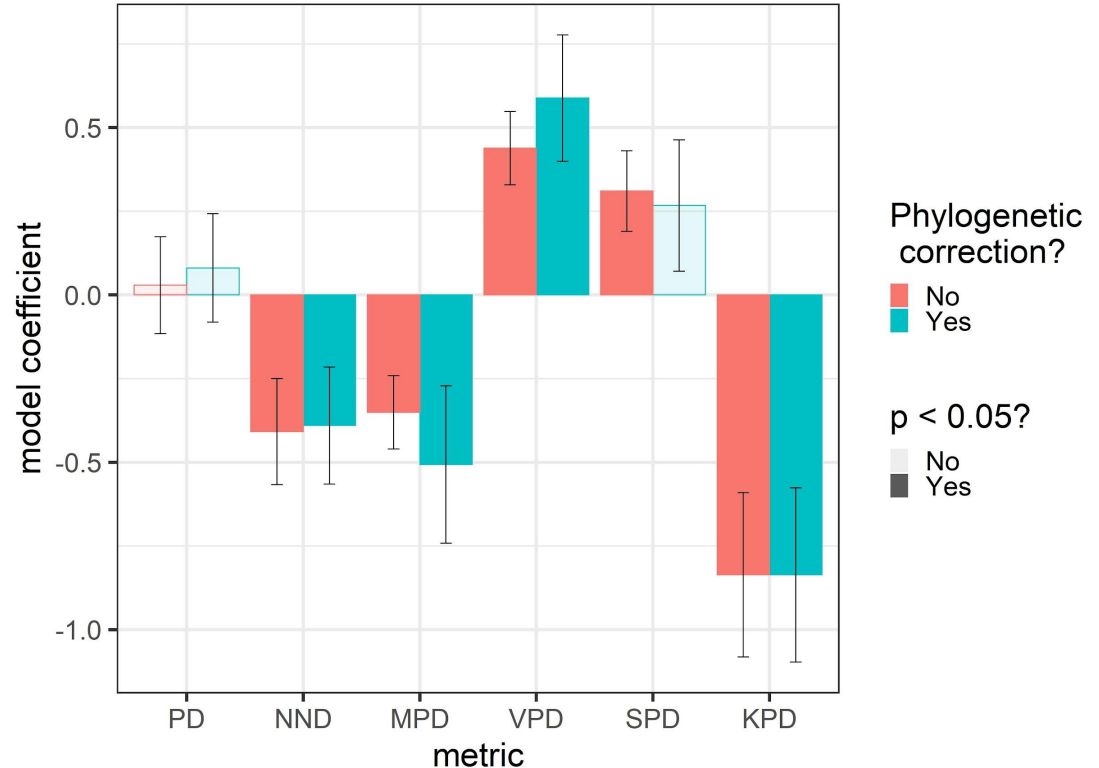
Used 6 metrics of phylogenetic structure

- Phylogenetic Diversity (PD; sum of branch lengths)
- Nearest Neighbor Distance (NND; minimum distance)
- Mean Phylogenetic Distance (MPD)
- Variance of Phylogenetic Distances (VPD)
- Skewness of Phylogenetic Distances (SPD)
- Kurtosis of Phylogenetic Distances (KPD)

Results: Does Source Region Matter?

Yes!

Source region community
phylogenetic structure
significantly associated with
establishment success



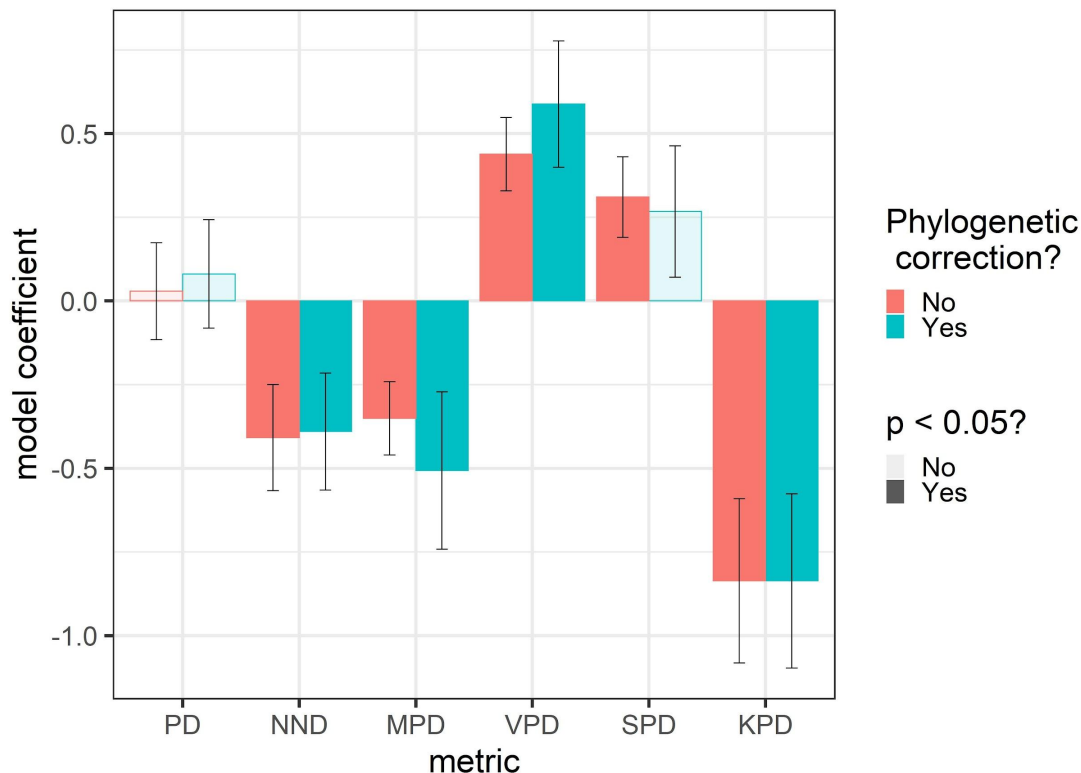
Results: Does Source Region Matter?

Yes!

Source region community phylogenetic structure significantly associated with establishment success

Results support EIH:

Species co-occurring with close relatives are more successful establishers



Analyses: Source and/or Recipient region?

Compared the fits of models containing:

- 1) Source region phylogenetic metrics
- 2) Recipient region phylogenetic metrics
- 3) Source AND recipient region phylogenetic metrics

Establishment success ~ source and/or recipient region phy. metrics +
source region species richness +
source region area + (1|recipient region)

Recipient region delineated 3 ways:

Native species only
Established species only
Native AND established species

Results: Source and/or Recipient region?

Best overall model ($\Delta AIC > 3$) included:

Negative effect of source region mean phylogenetic distance

Positive effect of recipient region (natives only) mean phylogenetic distance

Source region phylogenetic structure sometimes more important

Source region coefficients were consistent in sign across models

Recipient region coefficients varied in sign across models

Results: Source and/or Recipient region?

In other words:

- Successful species

 - Come from regions with close relatives

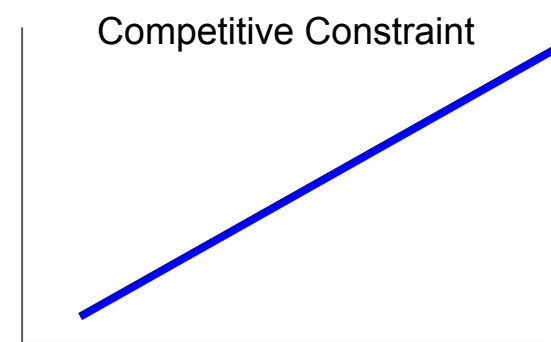
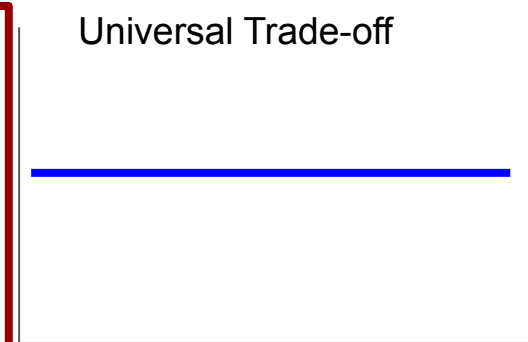
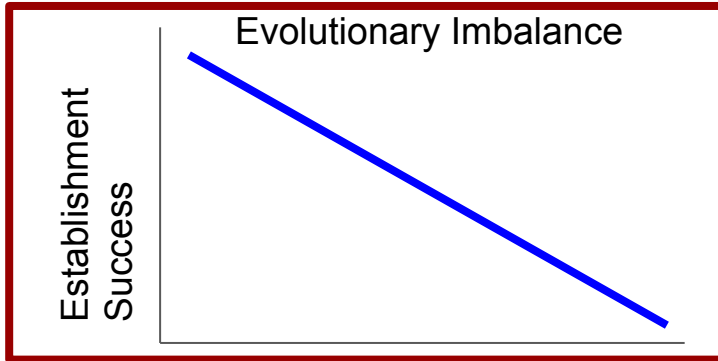
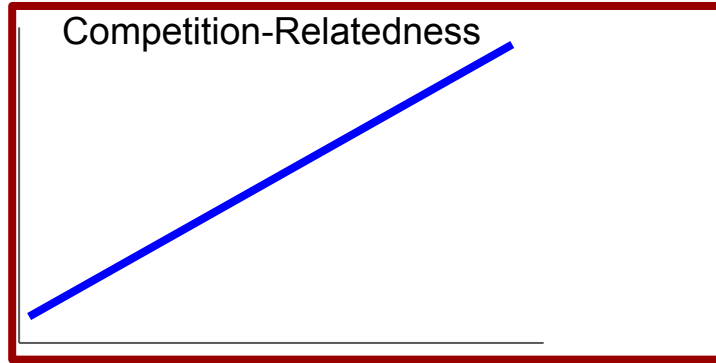
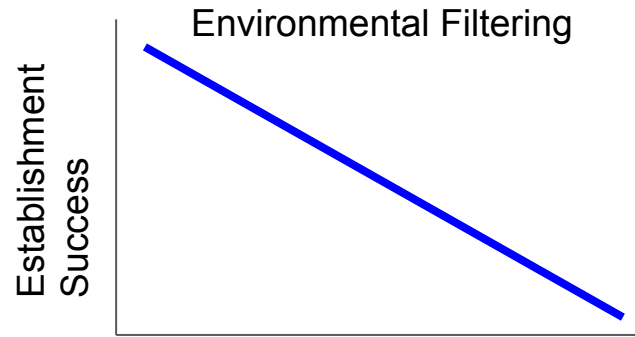
 - Are introduced into regions with distant relatives

- Consistent with an effect of competition/apparent competition

 - Supports EIH and CRH

- Source region phy. structure important

Establishment Hypotheses



Phylogenetic distance to **source** community

Phylogenies and establishment: Questions

Can source-range phylogenetic structure predict establishment success?

Yes! Sometimes better than recipient-region phylogenetic structure

Does incorporating source-range phylogenetic structure improve models?

Yes!

Which hypotheses are supported ?

Evolutionary Imbalance, Competition-Relatedness

Conclusions: Source region matters

Source region phylogenetic structure:

- Has been virtually ignored in the literature

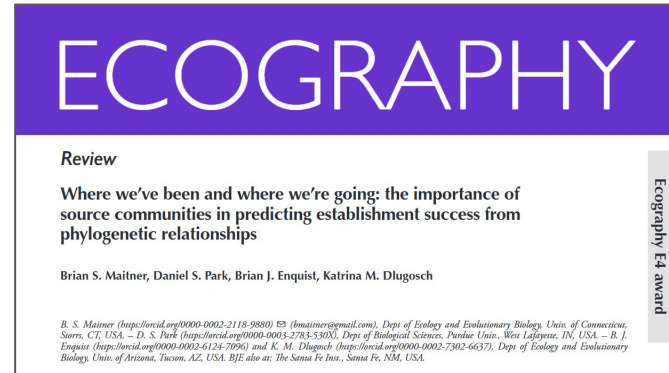
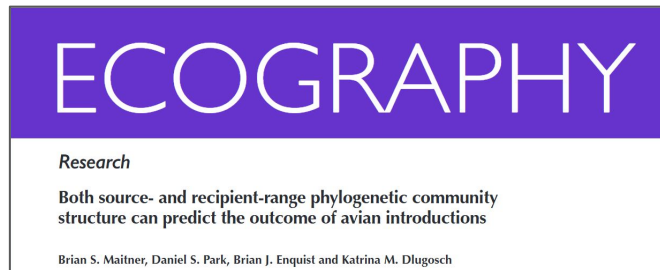
- But is significantly associated with establishment success

- Can improve models already containing recipient region phy structure

- Is often a stronger predictor than recipient region phy structure

These results suggest a need for more work on source-region effects in invasion

- May be useful to revisit previous work



Examples for today

- 1) Changing bird communities due to introductions
- 2) Changing communities due to climate change

Community assembly and climate

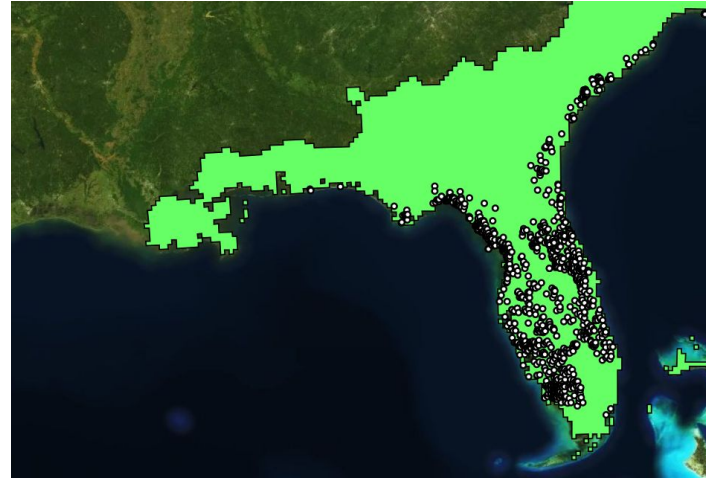
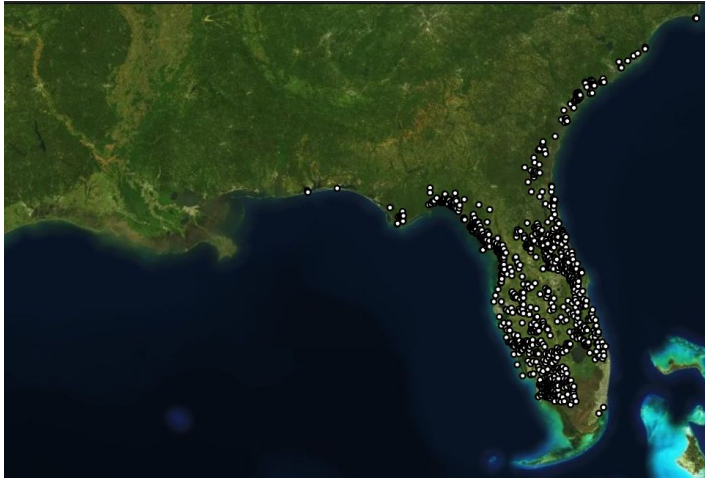
Distribution Modelling

- Usually incorporates abiotic factors
- In some cases, can also account for biotic interactions and dispersal

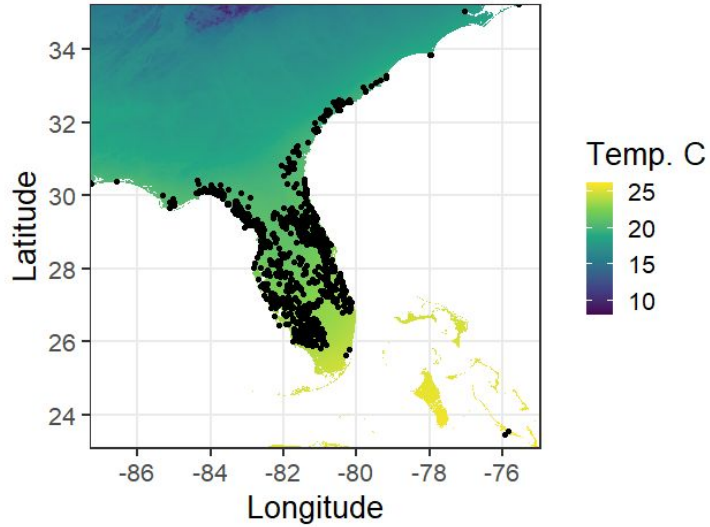
From Occurrences to Ranges



From Occurrences to Ranges

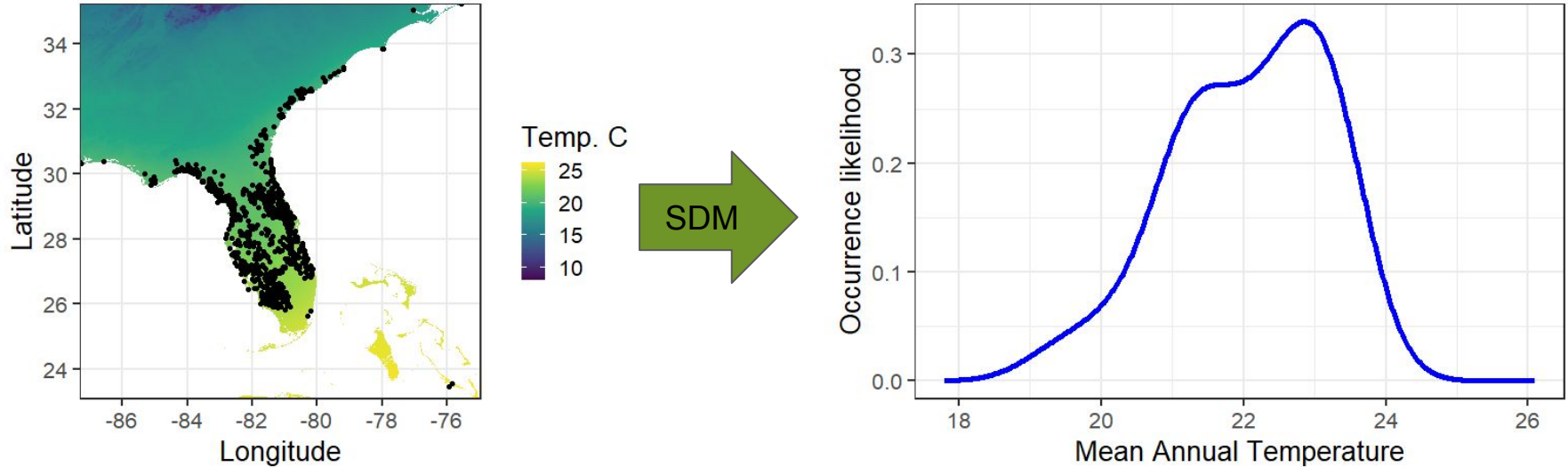


Distribution Modeling (a.k.a. Niche Modeling)



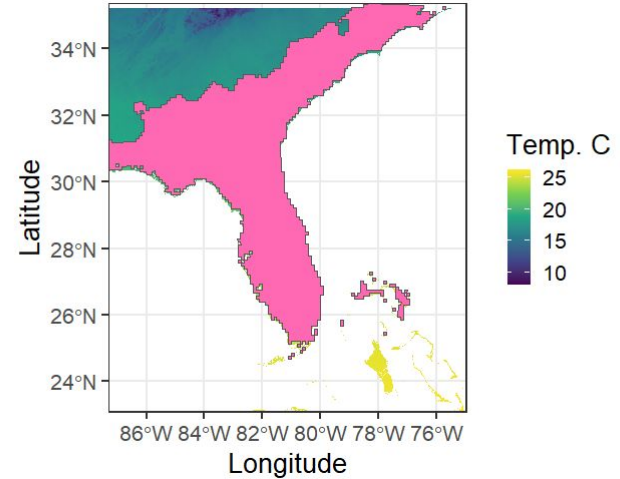
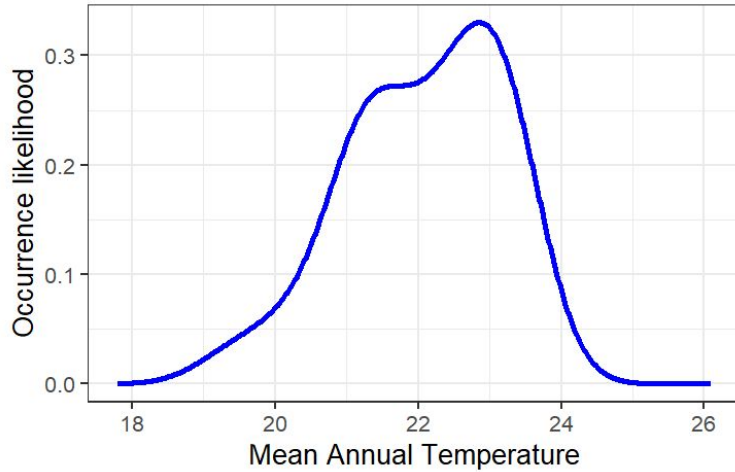
See: <https://wallaceecomod.github.io/>

Distribution Modeling (a.k.a. Niche Modeling)



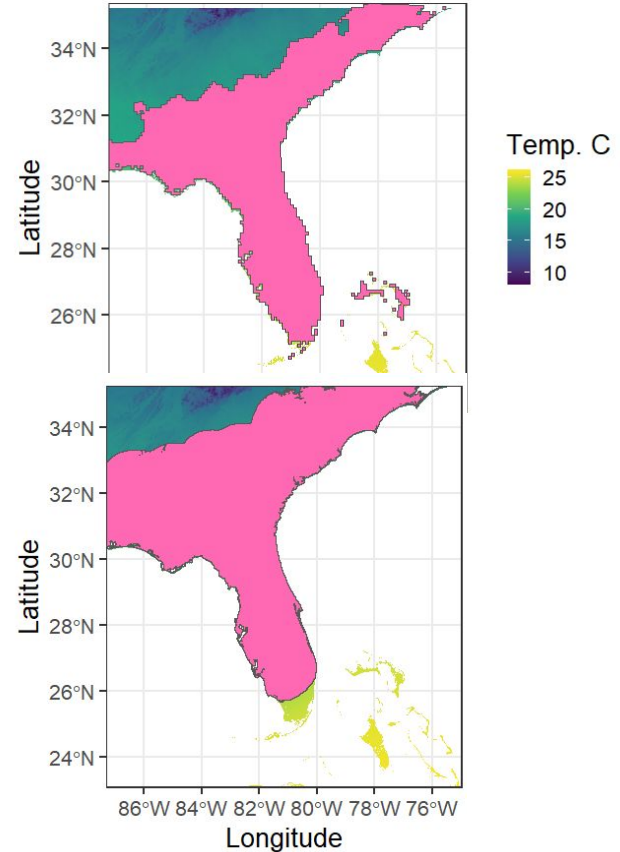
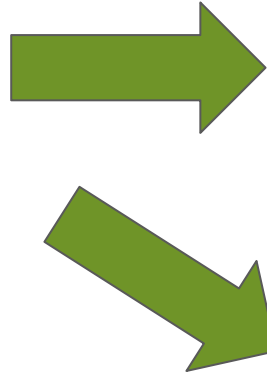
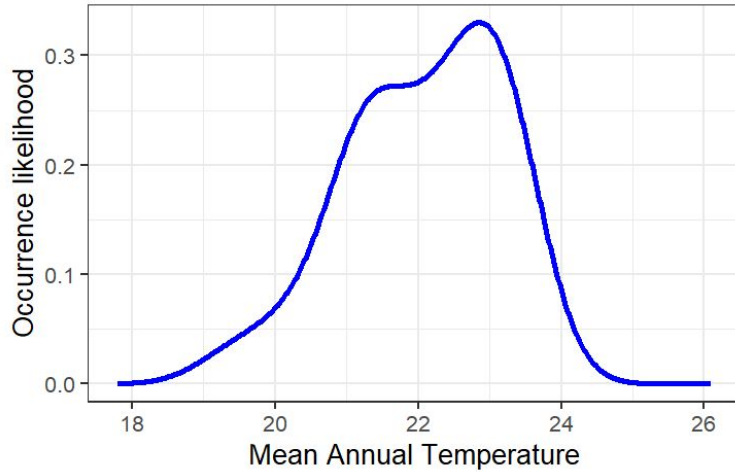
See: <https://wallaceecomod.github.io/>

Distribution Modeling (a.k.a. Niche Modeling)



See: <https://wallaceecomod.github.io/>

Distribution Modeling (a.k.a. Niche Modeling)

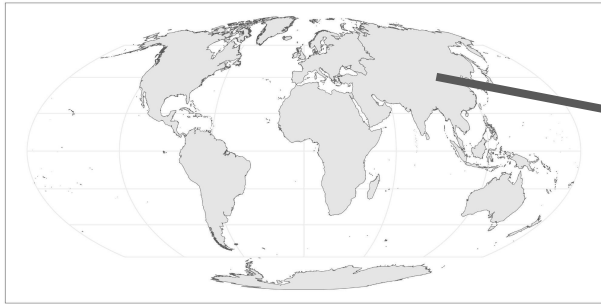


See: <https://wallaceecomod.github.io/>

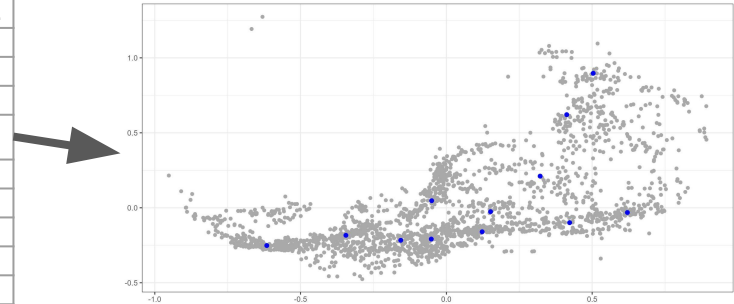
Species to communities

If we model lots of species, we can predict the communities, etc.

- This is useful if we want to focus on a particular area
- If we want to understand broad trends, we may want to aggregate similar communities into ecoregions

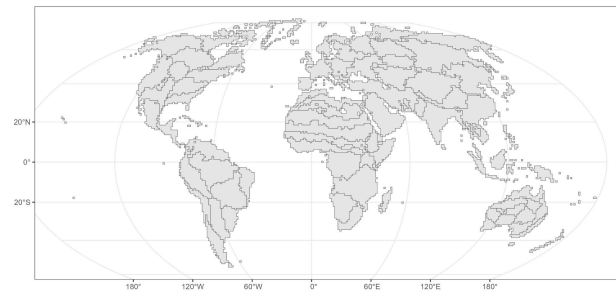
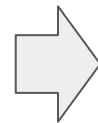
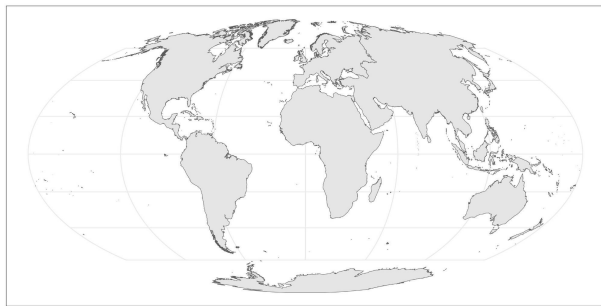


	s1	s2	s3	s4	s5	s6	s7	s8	...
c1	1	1	1	.	.	.	.	.	.
c2	.	1	.	.	.	1	.	.	.
c3	.	1	.	1	1	1	.	.	.
c4	.	.	.	.	.	.	1	.	.
c5	.	.	1	.	.	.	.	.	.
c6	.	.	.	.	1	.	.	.	.
c7	.	.	.	.	.	.	.	1	.
c8	.	.	.	1	.	.	.	.	1
...	1	.	.	.	.	.	.	.	.



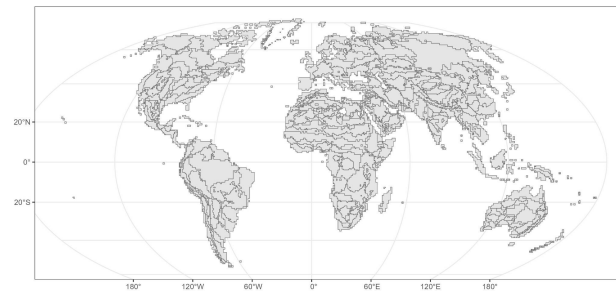
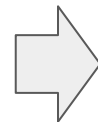
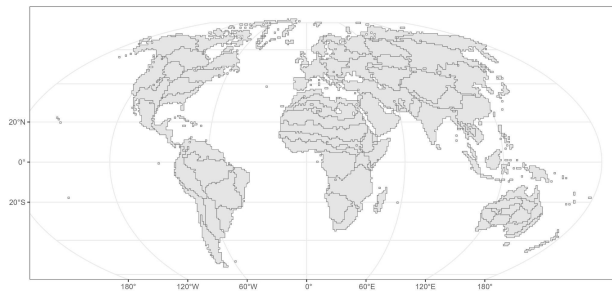
Step 1:

Assemblages
within continents
clustered into
level-0 ecoregions



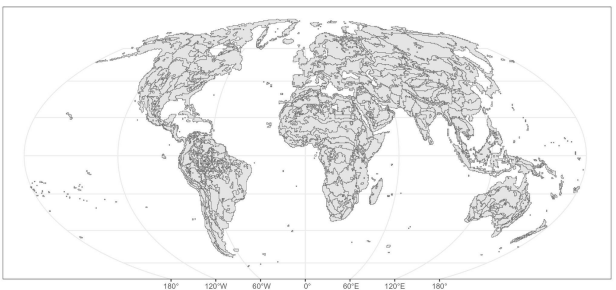
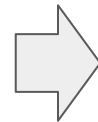
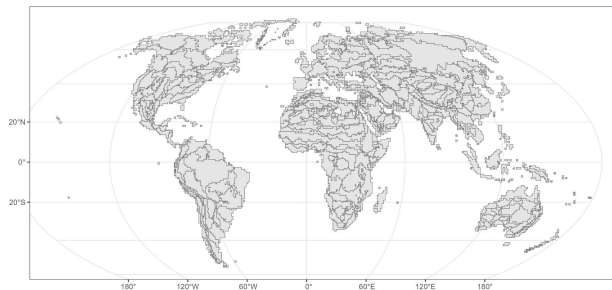
Step 2:

Assemblages
within level-0
ecoregions
clustered into
level-1 ecoregions

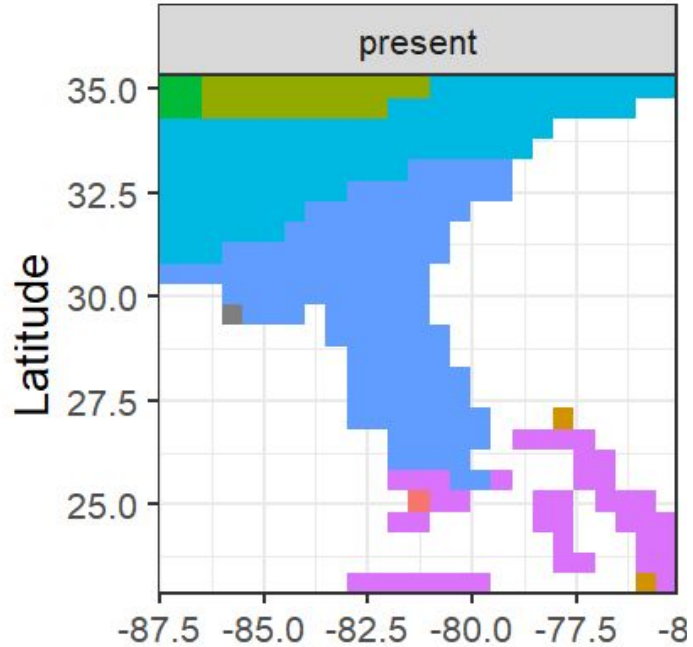


Step 3:

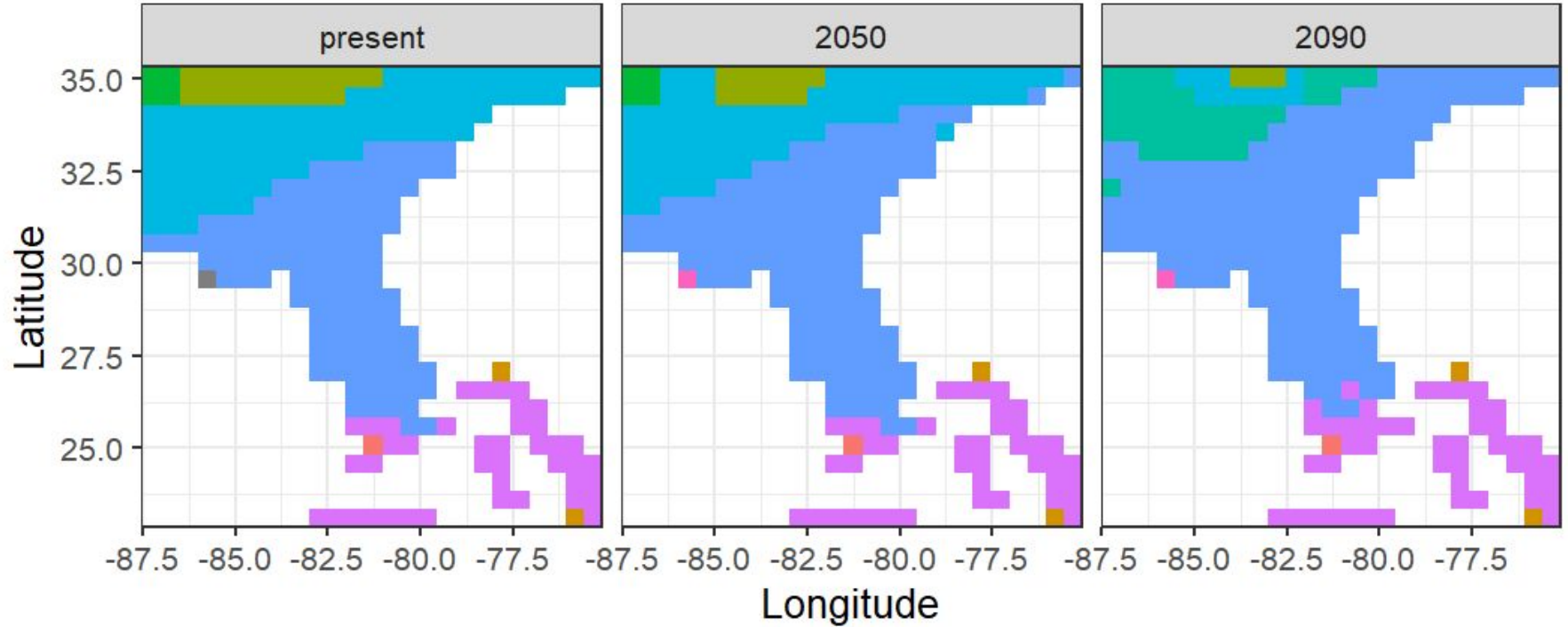
Level-1 ecoregions
projected into the
future



Combining many species into ecoregions



Ecoregions changing over time



Predicted community changes and conservation

Where should we preemptively protect?

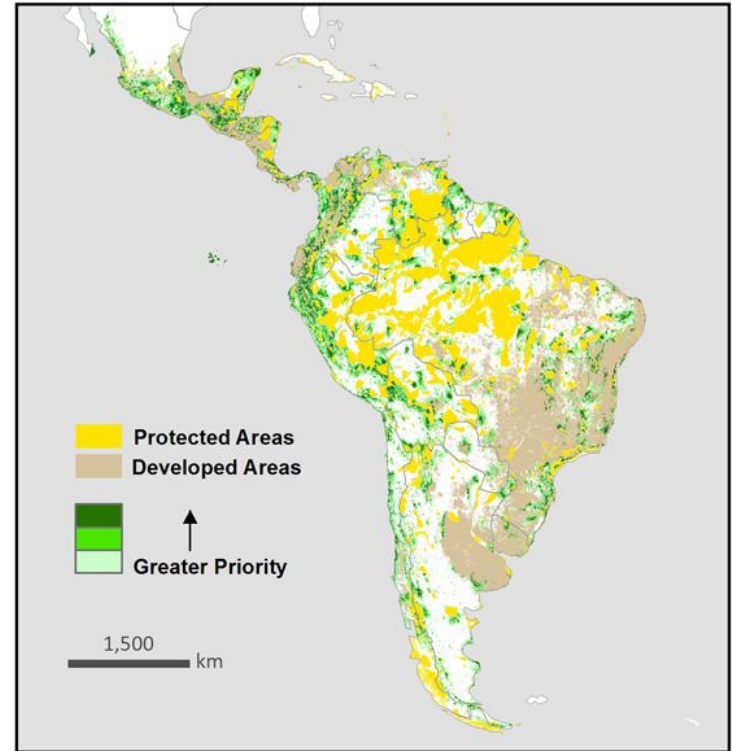
- Future hot spots
- Dispersal corridors



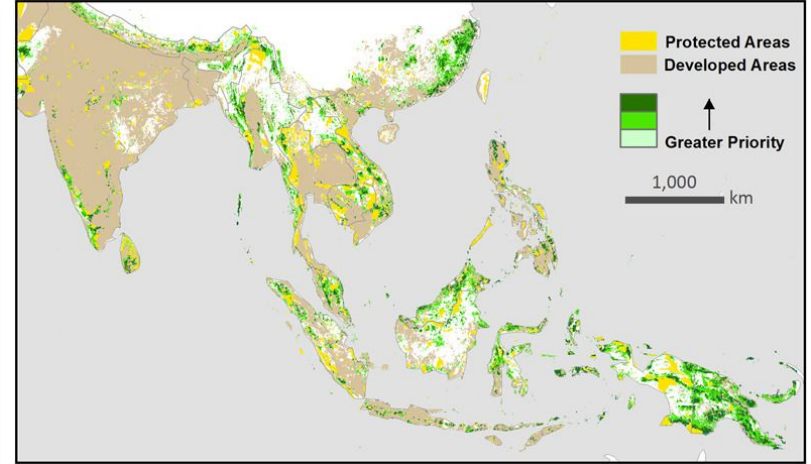
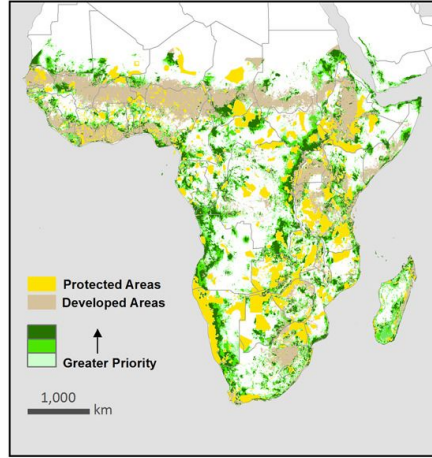
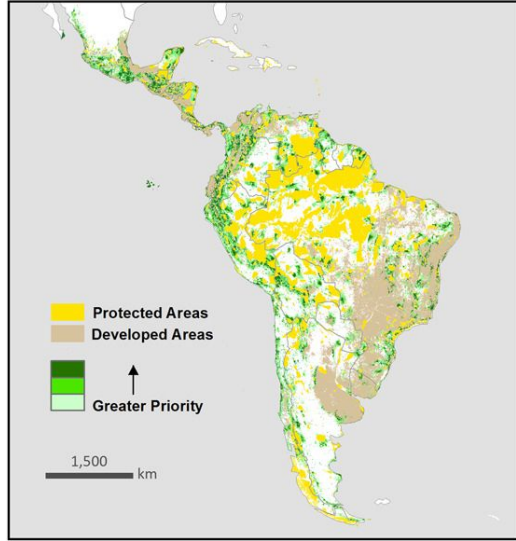
A JOURNAL OF SPACE
AND TIME IN ECOLOGY

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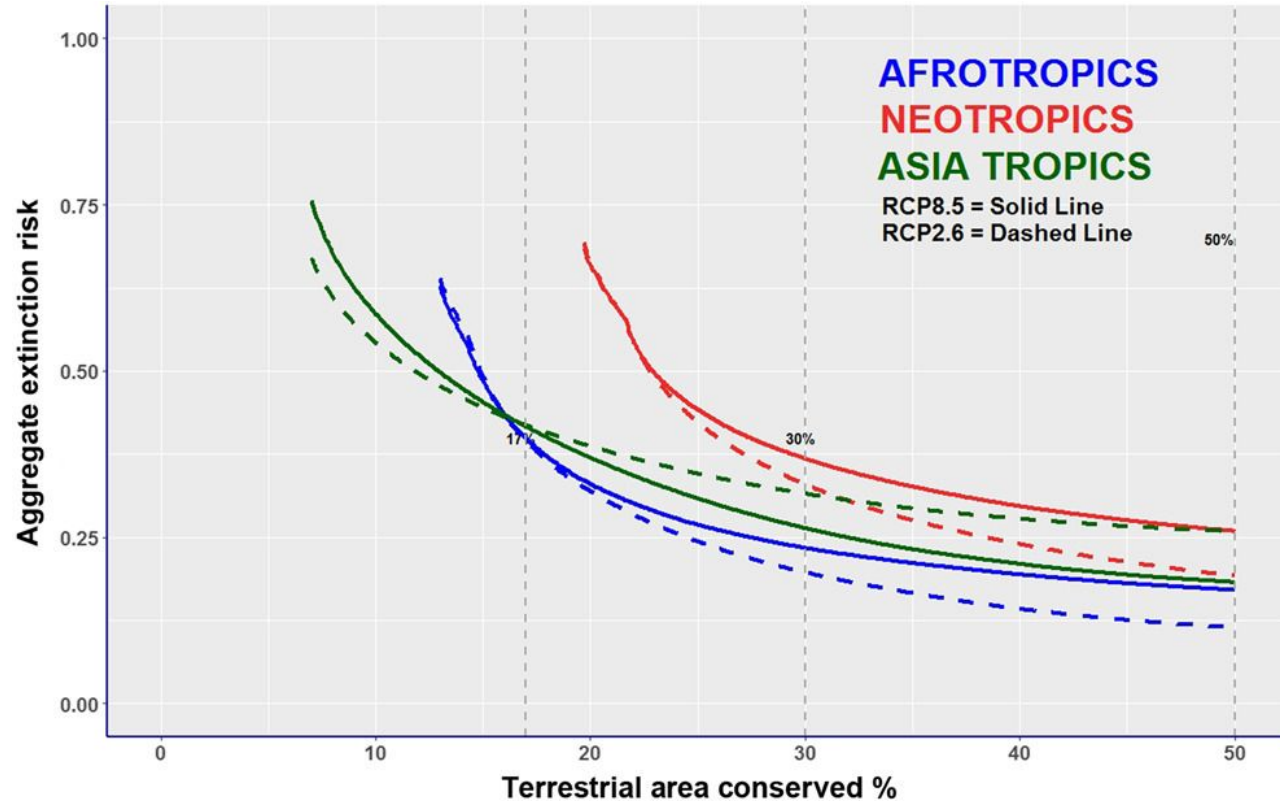
30% land conservation and climate action reduces tropical extinction risk by more than 50%



Predicted community changes and conservation



Predicted community changes and conservation



Next class: Student community assembly presentations